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CFD INVESTIGATION OF AN ISOLATED SURGING WIND TURBINE ROTOR AND A SURGING WIND TURBINE ROTOR IN WAKE

by

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Thesis submitted to the University of Plymouth
in partial fulfilment of the requirements for the degree of

MSc Offshore Renewable Energy Engineering

Faculty of Science and Engineering
School of Engineering, Computing and Mathematics

September 2023

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Abstract

CFD Investigation of an Isolated Surging Wind Turbine Rotor and a Surging Wind Turbine Rotor in Wake

by Aleksandr Tsvetkov

Floating offshore wind is an emerging marine renewable energy technology expected to contribute to decarbonisation of electricity generation. Presently, levelised cost of energy of floating offshore wind is high, and efforts are needed to reduce risks and increase investor confidence in the technology. One of the ways of doing so is to increase the accuracy of power output prediction of a floating wind turbine, especially when located in a wake. This work uses computational fluid dynamics to simulate a wind turbine rotor and compare its performance in fixed and surging conditions in both isolated and in a wake scenarios. To account for the geometry movement, sliding mesh interface and morphing mesh techniques are combined. The results are first validated against the available experimental data from the UNAFLOW campaign. For fixed cases, the agreement in forces and wake velocity distribution was found to be good. For surging cases, while forces were predicted well, discrepancy in wake velocity deficit was observed. The CFD results showed that a surging rotor promotes faster wake recovery, which was not observed in the experiment. Simulations of the rotor in a wake showed that, due to faster wake recovery behind the surging rotor, mean power coefficient is higher for a floating turbine than for a fixed one. The conclusion might inform future wind farm layout development, allowing to position the turbines closer to each other, or increase power production by keeping them at the same distance.

Word count for the main body of the thesis: 14 700.

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Symbols and abbreviations

A	rotor swept area, m^2
C_p	power coefficient, $-$
C_t	thrust coefficient, $-$
D	rotor diameter, m
e	error, $*$
g	acceleration due to gravity, $m\ s^{-2}$
k	turbulence kinetic energy, $m^2\ s^{-2}$
n	rotor speed, s^{-1}
p	pressure, Pa
P	rotor power, W
Q	rotor torque, $N\ m$
R	rotor radius, m
rms	root mean square, $*$
t	time, s
T	rotor thrust, N
U, u	velocity, $m\ s^{-1}$
V	wind speed, $m\ s^{-1}$
y^+	non-dimensional wall distance, $-$
μ	molecular dynamic viscosity, $m^2\ s^{-1}$
ρ	density, $kg\ m^{-3}$
ω	specific turbulence dissipation rate, s^{-1}
ADM	actuator disk model
ALM	actuator line model
AMI	arbitrary mesh interface
AoA	angle of attack
BEM	blade element momentum theory
CFD	computational fluid dynamics
DES	detached eddy simulation
DNS	direct numerical simulation
EU	The European Union
FEW	finite element analysis

FOWT	floating offshore wind turbine
FVM	finite volume method
LE	leading edge
LES	large eddy simulation
LCoE	levelised cost of energy
MRF	moving reference frame
NREL	National Renewable Energy Laboratory
NSE	Navier-Stokes equations
PDE	partial differential equation
PIV	particle image velocimetry
PSD	power spectral density
PV	photo-voltaics
RAO	response amplitude operator
RMS	root mean square
RWT	reference wind turbine
SMI	sliding mesh interface
SST	shear stress transport
TE	trailing edge
TSR	tip speed ratio
UK	The United Kingdom of Great Britain and Northern Ireland
UNAFLOW	UNsteady Aerodynamics of FLOating Wind turbines
US	The United States of America

Acknowledgements

I would like to express my gratitude to my supervisor Dr Edward Ransley for his support during the project. Thank you for giving me just enough freedom to pursue my interests, while providing helpful feedback to guide me towards a completion of this dissertation.

This work would have been impossible without the access to the computational facilities of the High Performance Computing Centre, University of Plymouth (`https://www.plymouth.ac.uk/about-us/university-structure/faculties/science-engineering/hpc`) and to the ARCHER2 UK National Supercomputing Service (`https://www.archer2.ac.uk`) granted via the UK High-End Computing Consortium for Wave Structure Interaction (HEC-WSI, `http://hec-wsi.ac.uk`), supported by EPSRC (grant no. EP/X035751/1).

I would also like to thank my family for their long-distance support.

1 Introduction

1.1 Renewable electricity generation

Electricity accounts for a significant share of about 20% in the global final energy consumption (IEA, 2022b). It is expected that electrification will play a crucial role in tackling climate change by contributing to decarbonisation of heating and cooling, transport, and industrial processes (HM Government, 2021). As a consequence, future projections by IEA (2022b) show that electricity may account for up to $28 - 52\%$ of global final energy consumption in 2050, depending on a particular scenario. This increase is likely to be covered by renewables, which are considered to be more environmentally friendly and secure energy sources than fossil fuels. As a result, the proportion of electricity generated from renewable sources is expected to increase from 30% in 2022 to 45-60% in 2050, which will require an unprecedented increase in renewable energy installations (IEA, 2022a).

Recently, spurred by the energy crisis and energy security concerns due to Russia's invasion of Ukraine, many governments have issued new policies supporting renewable energy generation. The EU has rolled out The Green Deal Industrial Plan (European Commission, 2023) aiming to create a more favourable regulatory and economic environment for renewables. Similarly, the UK outlines its short- and long-term ambitions and policy in The British Energy Security Strategy (HM Government, 2022), the US - in its Inflation Reduction Act (The White House, 2022), and China - in its 14th Five-Year-Plan (United Nations Development Programme, 2021). Governments put great emphasis on solar PV and wind energy, and it is estimated that roughly 90% of all new energy installations will come from wind and solar (IEA, 2022a). Furthermore, economic analyses show that solar and wind remain the cheapest options for new electricity generation in the majority of countries (IEA, 2023), despite an increase in commodity prices due to Covid-19 pandemic and a war in Ukraine, which demonstrates resilience of the industry. It is fairly certain, therefore, that the interest in renewables will remain sustained.

Although solar PV is the cheapest option for electricity generation in the majority of countries worldwide (IRENA, 2022), wind energy is an expanding market still, bringing diversification of energy mix. Wind turbines can be installed either on land (onshore) or at sea (offshore). Onshore wind, being a predecessor of offshore wind, accounts for the majority of the total installed wind capacity: 836GW out of 906GW (IRENA, 2022; GWEC, 2023). Despite its higher levelised cost of energy (LCoE), offshore wind offers some advantages, such as access to greater wind resource, fewer restrictions to construc-

tion of ever larger wind turbines, and decreased visual and noise impacts, which allowed it to secure its rightful place in the energy mix. Many countries have established frameworks supporting planning and consenting, supply chains, market policies and included offshore wind installations as part of their energy strategy for the upcoming decades. For example, the UK targets 40GW of offshore wind capacity by 2030 (HM Government, 2021), Europe, within its REPowerEU framework, aims at 60GW by the same year (European Commission, 2020), and The Inflation Reduction Act in the US eyes 30 GW. As a result, offshore wind installations are expected to accelerate.

Within offshore wind technology, two sub-technologies can be pointed out: fixed-bottom and floating. They refer to the installation method, of which the fixed structures are rigidly secured in the seabed, while floating foundations provide support and stability via buoyancy and mooring. Currently, fixed foundations are the predominant offshore wind turbine installation technique. However, floating foundations allow for deployment in deeper waters, where fixed structures tend to be prohibitively expensive (Greaves et al., 2022). This will allow to access greater wind resource outside of relatively shallow offshore waters such as in the North Sea in Northern Europe and on the west and south-west coasts of China, where the majority of installations have been taking place so far. Further opportunities may be opened up for virtually any country with an access to offshore waters, including but not limited to Portugal, Italy, India, Japan, Brazil and the USA (GWEC, 2022).

At present, the floating offshore wind industry is at its early stage of development, which is characterised by a relatively high LCoE, lack of convergence in platform design and a low number of deployed full-scale prototypes (Edwards et al., 2023).Few farm-scale test projects have been installed so far, namely Hywind Scotland (*Hywind Scotland* 2023), WindFloat Atlantic (*WindFloat Atlantic* 2023), and Kincardine (*Kincardine Offshore Wind Farm* 2023). Nevertheless, ambitions are high to accelerate the deployment of floating offshore wind farms. In the most recent ScotWind leasing round, more than 17GW worth of capacity were awarded for floating offshore wind projects (Crown Estate Scotland, 2023). England eyes 5GW of floating offshore wind and is in the process of preparing the Celtic Sea Leasing Round (The Crown Estate, 2023). Similarly, more than 4GW are expected to be built off the coast of California (BOEM, 2023). Overall, it is reported that the global pipeline has reached 185GW (RenewableUK, 2022), although stages of development vary.

1.2 Modelling of floating offshore wind turbines

Ambitious targets in floating offshore wind installations manifest themselves in active research to address challenges and open up opportunities within the industry (Greaves et al., 2022). Although tackling financial, legislative and supply-chain constraints is of paramount importance to the success of the sector (IEA, 2023), this section will focus on technical issues. Pertinent to marine renewable energy in general, Greaves et al. (2022) summarise main challenges, which are concerned with (i) design and validation of the next generation technologies, (ii) grid integration, (iii) data collection, analysis and modelling tools, (iv) operation and maintenance strategies. Of particular interest are the modelling approaches, as they allow to test prototypes and support the deployment of full-scale projects.

Floating offshore wind turbines represent a complex system due to coupling of many disciplines, namely aerodynamics, hydrodynamics, structural dynamics, electrical and control engineering. Otter et al. (2022) give an overview of techniques used to study floating wind turbines. Two major approaches in modelling are the physical and numerical testing. Physical experiments investigate a real-life prototype of a system, albeit usually at a smaller laboratory scale. They allow to observe the actual phenomena of interest under specified conditions and collect valuable performance data, which are needed for both design and validation of numerical tools. Physical experiments are an indispensable tool for researchers but have their disadvantages. They require big state-of-the-art facilities and expensive and time-consuming campaigns. Scale laws introduce uncertainty to the extrapolation of results to full scale, which significantly affects the accuracy. Multidisciplinary nature of the floating wind turbines further increases the complexity of an experimental setup (Subbulakshmi et al., 2022).

An alternative to the physical modelling is the numerical modelling. Physical phenomena can be expressed in terms of mathematical models of varying fidelity. Depending on the nature of a problem, various assumptions can be made to simplify the mathematical description of the problem, such as using a potential flow theory to neglect viscous terms, considering laminar flow to avoid dealing with turbulence, assuming the flow to be incompressible to reduce the number of variables. The governing equations are discretised into systems of algebraic equations, which can be solved with the help of widely available computing power. A large number of models have been developed over the years to model different processes. The choice of the model is usually a compromise between its accuracy and computational intensity (Otter et al., 2022). Low fidelity approaches (those which use

"simpler" models) allow to obtain results quickly, facilitating optimisation studies with a large number of test cases. High fidelity approaches (those that use more "sophisticated" models), on the other hand, offer more precision at the expense of computational time. Generally speaking, when compared to physical modelling, numerical experiments can be cheaper, faster, allow to obtain results for a wider range of input conditions and can give extensive information about the solution. The disadvantage of numerical modelling is that it effectively investigates the mathematical model rather than the actual phenomena, which introduces uncertainty to the final results. To establish confidence in numerical models, they must be carefully verified and validated before being used for engineering purposes and then continuously validated against newly available data.

Aerodynamics of a wind turbine is one of the main drivers behind its efficiency. A wind turbine blade is essentially a wing which generates lift in a rotor plane resulting in torque and rotation of the rotor. Aerodynamic lift is generated due to specific motion of a fluid around an object. To estimate the lift, it is necessary to determine the characteristics of the flow field. Many models have been introduced to tackle this problem. The works of Sanderse, Van Der Pijl, and Koren (2011) and Subbulakshmi et al. (2022) give an overview of methods widely used in the prediction of wind turbine aerodynamic characteristics. The three big families of methods are the blade element momentum theory (BEM), vortex-based methods, and computational fluid dynamics (CFD).

Blade element momentum theory combines a simplified description of rotor blades and the conservation of momentum principle. A rotor is discretised into annular ring elements with their corresponding 2D lift and drag coefficients, which can be derived from blade aerofoil performance data. The force on each section can be written using known inflow speed, which is a combination of far-field wind speed and rotor rotation. The estimated forces are then assumed to be responsible for the change in momentum of the fluid. Writing the conservation of axial momentum equation allows to obtain the governing equation for the problem and consequently a way to estimate the required forces.

BEM is a popular engineering tool in wind energy industry and is widely used for the design of onshore and fixed offshore turbines. Matha et al. (2011) point out, however, that care must be taken when using the method for floating wind turbines, as BEM assumes the flow to be strictly axial, neglects radial velocity components, and does not model the wake explicitly. A turbine on a floating foundation is likely to experience large motions, leading to non-axial velocities and interaction with its own wake, which cannot be captured by the model. Although BEM limitations prompted the development of various correction models, such as tip-loss correction (Oliveira et al., 2019), further investigations are needed

to assess the accuracy of the method in unconventional flow regimes.

Another family of methods is the so called vortex methods. These models discretise rotor blades with "elementary flows" (Anderson, 2017), vortices, and then compute the flow field and forces based on the Biot-Savart law and the Kutta-Joukowski theorem. These methods are less widespread than BEM and CFD. However, Ramos-García, Kontos, et al. (2022) have extensively tested their in-house lifting line code and claim that it is a very promising approach as it allows to obtain more information about the flow than BEM, but at a lower computational effort than CFD.

CFD is a general method for investigating fluid flows and can be applied to study wind turbines. CFD attempts to solve the Navier-Stokes equations (NSE), the governing equations of fluid flow, by discretising them in space and time. Unlike previously mentioned methods, CFD can handle less strict assumptions about the fluid flow and therefore tends to be better at capturing viscous and non-linear phenomena and is generally considered to be a high-fidelity method, allowing to obtain extensive information about the fluid flow. CFD is a computationally intensive technique and is known to be more of a research tool rather an engineering one. It is quite often considered for investigating new problems, such as moving wind turbines on floating foundations (Tran and Kim, 2016; Cormier et al., 2018).

In CFD, it is common to introduce some simplifications, because a complete modelling of all flow details of a real-life engineering problem tends to be too computationally expensive. For example, the rotor can be discretised as a disk with constant or varying force distribution, which is known as the Actuator Disk Method (ADM) (Sanderse, Van Der Pijl, and Koren, 2011). Conversely, each blade can be depicted as a line, which gives the foundation for the Actuator Line Method (ALM). Further, the blade can be represented by a flat surface with force distributed over it. Finally, a fully resolved simulation considers actual 3D blade geometry. Another aspect of the simulation which requires additional consideration is the choice of a turbulence model. Turbulence is a complex phenomenon and is not yet fully understood. In the order of decreasing detail captured by the model, the existing approaches are a direct numerical simulation (DNS), large eddy simulation (LES), and Reynolds-averaged Navier-Stokes equations (RANSE) (Versteeg and Malalasekera, 2007). The choice of an appropriate modelling approach is one of the challenges of CFD.

Moving on from turbine scale to wind-farm scale, an important part of the flow which needs to be resolved is a wake behind a turbine. The wake is a region downwind the turbine characterised by reduced wind speed and increased turbulence (Crespo, Hernández,

and Frandsen, 1999). Wakes affect downwind turbines by decreasing their output and increasing structural loading. Modelling the wake is particularly challenging because it depends on numerous factors, such as wind speed, wind shear, atmospheric turbulence, and particular parameters of a turbine, such as aerofoils, blade pitch, rotor yaw misalignment, and turbine motion in the case of a floating design.

Similar to wind turbine aerodynamics, models of varying fidelity are available to account for wakes. Analytical wake loss models are widely used and depend on few integral parameters such as principal dimensions and thrust coefficient. Due to their computational simplicity, these models can be used on wind farm scale and are shown to have been calibrated to be quite accurate (Archer et al., 2018). However, since they were calibrated specifically for fixed turbines, they are not directly applicable to the floating turbine wakes. Due to the low number of prototype installations and complexity of measurement campaigns, experimental data covering floating wind turbine wakes is not readily available. As a result, numerical models tend to be used instead of analytical ones. Among those, CFD relies on fewer assumptions about the flow than other models. In particular, it can consider important contributions due to viscosity and turbulence. It is also a high-fidelity tool and, therefore, tends to be the method of choice for detailed investigations of these new operating conditions (de Vaal, M. Hansen, and Moan, 2014; H. Lee and D.-J. Lee, 2019; Rezaeiha and Micallef, 2021).

Fixed offshore wind turbines have been studied extensively over the last decades, which enabled the development of various models for prediction of energy loss due to wake effects (Archer et al., 2018). With the emergence of floating foundations, however, the turbines are expected to experience 6-degree-of-freedom motions not characteristic for the fixed ones. It is suggested by Sebastian and Lackner (2013) that the already complicated problem of wind turbine aerodynamics becomes even more complex. At present, to make the problem more manageable, it is common to investigate one motion of the turbine at a time (H. Lee and D.-J. Lee, 2019). It was identified that motions with the highest axial velocity component, i.e., surge and pitch, affect the turbine performance the most.

It is natural to expect for floating wind turbines to be installed in large arrays, similar to onshore and fixed offshore wind turbines. An important consideration in increasing the efficiency of such installations is the problem of aerodynamic interaction of machines which are located relatively close to each other. Some works suggest that rotor oscillations in all directions have an effect on the wake structure, introducing periodic variations which promote the rapid breakdown of the vortical structures (Kopperstad, Kumar, and Shoele, 2020; Rezaeiha and Micallef, 2021). The resulting faster wake recovery might be exploited

to inform a more tight wind farm layout. The current project aims to contribute to the research in this area. Given that the aerodynamics of floating wind turbines is a relatively new field of study, high-fidelity methods are a preferred tool, as they avoid making any severe assumptions about the fluid flow, the effect of which is not yet fully understood. Consequently, CFD is used in this project.

1.3 Aims and objectives

The proposed project aims to investigate the influence of a surging motion of a wind turbine rotor on its power production, as well as on power production of a downstream turbine. The identified objectives are as follows.

1. Choose an appropriate numerical tool, which can reliably capture phenomena of interest.
2. Select a turbine, preferably the one that has been studied before in order to enable validation of results.
3. Decide on the representative test case scenarios.
4. Investigate the implementation of mesh motion.
5. Investigate the implementation of wake interaction.
6. Carry out the numerical simulations and interpret the results.

1.4 Report structure

The report is organised as follows. Firstly, section 2 gives an overview of the research landscape in the field of aerodynamics of floating offshore wind turbines. Next, the methodology of the study is discussed in section 3, with the focus on the numerical setup. Results are then presented and discussed in section 4. Finally, conclusions are summarised in section 5.

2 Research context

One direction of research in offshore renewable energy is experimental and numerical modelling, as these help to understand the behaviour of new systems and inform their design. A review of existing approaches to modelling floating offshore wind turbines is given by Subbulakshmi et al. (2022) and Bashetty and Ozcelik (2021). The authors emphasize that the response of the whole system is subject to strong coupling of aerodynamics, hydrodynamics, structural dynamics, servodynamics, and mooring dynamics. It is therefore challenging to conduct controlled experiments of the whole system, and research tends to focus on one or few phenomena, while employing various assumptions about the remaining ones. For example, the works of Wang et al. (2022) and Clement et al. (2022) investigated motion parameters of a floating substructure, while the studies of Porté-Agel, Bastankhah, and Shamsoddin (2020) and Brusca et al. (2018) attempted to explore aerodynamics in greater detail.

Problems associated with floating offshore wind are very complex, and collaborative effort is required to tackle them. Various joint projects, such as the IEA Wind TCP Task 30 (IEA Wind TCP Task 30, 2019) and UNAFLOW (Bayati, Belloli, Bernini, Boldrin, et al., 2018) have been launched to systematically investigate the principles behind operation of floating offshore wind turbines. Studies of turbine aerodynamics are crucial in increasing the overall efficiency of wind turbine systems and will be the focus of this project. Published works can be loosely classified as belonging to two types, the ones which primary aim is to characterise the response of a floating (i.e., moving) wind turbine rotor and the ones which aim to compare various models between themselves or with experimental measurements. The focus of the investigation tends to be on rotor integral forces and velocity distribution in the wake.

2.1 Early studies

Early studies into aerodynamic response of a floating wind turbine are characterised by building on the knowledge obtained in the research and design of fixed wind turbines. Furthermore, they tend to use simpler numerical models, such as BEM and actuator disk as opposed to, for example, fully resolved CFD.

Sebastian and Lackner (2013) admit that the aerodynamics of a fixed turbine is already sophisticated, but emphasize that additional degrees of freedom of movement of floating offshore wind turbines create an even more complex flow field. They summarise that the main contributors to this are the non-axial flow through the rotor disk, variation of wind

speed along the height of the rotor, and the additional wind component due to motions of a structure in waves. The authors further elaborate that, in these highly transient flow regimes, the assumptions made in the established models may be violated, leading to inaccurate predictions of velocities in the wind turbine wake and, consequently, energy yield of downstream turbines. It is important, therefore, that existing approaches are validated.

The works of de Vaal, M. Hansen, and Moan (2014) and Micallef and Sant (2015) compared the performance of the actuator disk model and the blade element momentum method. The results suggest that the transient phenomena can be captured qualitatively by both models, however the accuracy tends to depend on the motion parameters of the system. In particular, Micallef and Sant (2015) conclude that the best results are obtained when mean quantities are required or when the tip speed ratio and wave amplitudes are small.

One of the first experimental studies of an oscillating rotor was conducted by Farrugia, Sant, and Micallef (2014) with further analysis in Sant et al. (2015). The aim of the project was to gain a deeper understanding of unsteady aerodynamics of a turbine subject to surge motion. The authors investigated blade forces, averaged rotor forces, and wake. The results show that power characteristics of a surging rotor fluctuate considerably. Authors suggest that these fluctuations are crucial for the design from the point of view of fatigue loading and blade control strategies. However, they have identified a limitation of their experimental setup, as the torque sensor was not able to depict the torque variations due to rotor surge, despite correctly measuring the mean value. This further highlights that FOWT experiments should be carefully planned.

One of the first studies of a floating wind turbine using blade-resolved CFD was conducted by Tran and Kim (2016). The work focused on investigating flow parameters around a blade in unsteady conditions. It was concluded that, indeed, the aerofoils experience a change of angle of attack, which leads to variations in lift and drag, and, consequently, thrust and torque. Additionally, a comparison of rotor integral characteristics (thrust and torque) was carried out. The results between BEM and CFD methods were qualitatively similar, although differences in numerical values were observed. Since BEM does not model the wake, its investigation was based solely on CFD results. The authors observed that blade tip vortex changes its pattern due to constantly varying shedding position. They further speculated that this might have consequences on the far-wake velocity deficit.

Tabib et al. (2017) studied an NREL 5MW reference wind turbine (RWT) using three

approaches, the actuator line model (ALM), the sliding mesh interface (SMI), and the moving reference frame (MRF). The predicted power coefficient values were scattered, and SMI was closer to the available data than ALM and MRF. The results appeared to agree qualitatively, which is in line with the findings of Tran and Kim (2016). However, an error of up to 20% was observed. As for the wake, the higher fidelity SMI approach expectedly predicted a more complex 3D flow.

H. Lee and D.-J. Lee (2019) simulated a wind turbine rotor undergoing prescribed motions in each of the 6-degree-of-freedom directions separately using a vortex method. It was concluded that, out of all 6 motions, the stream-wise motions of surge and pitch affect the thrust force and power output most significantly as they contribute to the change of effective angle of attack the most. Meanwhile, oscillations in all directions had an effect on the wake structure, introducing periodic variations which promote rapid breakdown of the vortical structures. It was further pointed out that, because various types of platforms have different response in waves, the wake structure in each case varies as well.

To further highlight the differences in response of different substructures, Kopperstad, Kumar, and Shoele (2020) compared floating wind turbines installed on a spar-type and barge-type floaters. The turbines were modelled by an Actuator Disk Model (ADM), while the wake was resolved with LES. Different floaters were considered based on their characteristic response amplitude operators (RAOs) in similar wave climates. The results suggest that the turbine on the spar floater experiences oscillations of lower magnitude, which might be beneficial from the point of view of fatigue loading. The turbine on the barge, however, oscillates more which promotes faster wake breakdown, which in turn might be beneficial when turbines in a farm arrangement are considered.

In summary, the early studies show that, indeed, a FOWT experiences a high variation in its performance parameters, namely thrust and torque, which can be captured by the available aerodynamic models. Furthermore, it appears that the additional mixing in the wake due to the rotor motion, which can only be predicted by vortex methods and CFD, promoted faster wake recovery. All of these features seem to be important to quantify accurately to inform the studies of blade structural response, control strategies, and energy yield. However, the lack of reliable experimental results did not allow to make any judgement on the accuracy of the available methods.

2.2 The UNAFLOW project

To address the gap in available experimental data on oscillating wind turbines, a UNAFLOW (UNsteady Aerodynamics of FLOating Wind turbines) project was carried out. Bayati, Belloli, Bernini, Boldrin, et al. (2018) undertook an extensive experimental campaign aimed at evaluating the aerodynamic response of a model surging and pitching wind turbine. The turbine model was developed based on performance scaling, which meant reproducing the thrust and power coefficient of the DTU 10MW reference wind turbine on a one-third-scale wind (Bayati, Belloli, Bernini, and Zasso, 2017). It is claimed that ensuring constant thrust and torque coefficients between a prototype and a model is required to properly represent dynamic coupling between components of a floating wind turbine. This required the development of new blades with new aerofoils, chord and twist distribution. Further details on the experimental setup can be found in Bayati, Belloli, Bernini, Giberti, et al. (2017). The challenges noted by the authors are the structural and aeroelastic response of the turbine tower and blades and ensuring that they do not interfere with the thrust and torque measurements.

During the experiment, rotor integral aerodynamic loads were measured, together with hot-wire anemometry for wake deficit at certain positions downwind the turbine and particle image velocimetry (PIV) for evaluating vortical structures. The results are presented in Bayati, Bernini, et al. (2018) and further analysed in Fontanella et al. (2021). It was determined that, at low reduced frequencies below 0.5 Hz, the behaviour of the system can be predicted well by a quasi-steady theory, in other words, unsteady effects were negligible. However, the authors mention that the experimental frequencies were chosen in a way that ensured that the variation of the angle of attack (AoA) on the blade aerofoils remained within the linear region of lift coefficient curve. This might have allowed the quasi-steady theory to show good agreement with the experiment. It is uncertain, however, whether such a state is representative of the actual operational conditions. Nevertheless, at high reduced frequencies above 0.5 Hz the discrepancy was more apparent, which might indicate a bigger role of the unsteady effects in the overall rotor response.

Regarding the wake, interestingly, the average velocity deficit at a chosen distance downwind the turbine appears to be very similar in both fixed and surging cases. This conclusion contradicts the results of previously mentioned early studies but might have been the result of the wind tunnel blockage effect interfering with the wake propagation. Nevertheless, the pattern of vortical structures was clearly modified by the surging motion,

which was evident from the relative position of the tip vortex to the fixed reference frame and from power spectral density of axial velocity, which had peaks at the frequency of the surge motion. Analysis of PIV flow visualisation showed that energy content in the wake becomes smeared over wider frequency band, peaking at the surge frequency. The blade tip vortex also changes its travel speed, following the periodical oscillations of the turbine.

The UNAFLOW project provided an extensive dataset, which was used in a number of subsequent studies, primarily for validation of various numerical models. For example, Cormier et al. (2018) compared numerical results of the BEM theory, a RANS solver, and a vortex-based method. Incidentally, the RANS method used an overset mesh technique and modelled 1/3 of a turbine (i.e., one blade only) making use of the simplification based on the cyclic nature of the problem. The comparison revealed that all methods predict the thrust fairly well with errors about 3-4%. Torque, on the other hand, showed worse agreement with errors reaching 10-12%. The unsteady results captured the overall response well, but the thrust and torque were scattered within $\pm 10\%$. Interestingly, BEM theory was the most accurate method, which might indicate the need for the review of numerical setups of other methods. As noted by the authors themselves, "further investigations are currently undertaken to understand the reasons for this."

Similar to the previous study, Mancini et al. (2020) compared results from various models, including CFD, BEM and a newly developed quasi-steady approach. The comparisons suggest that, at low oscillation frequency, the quasi-steady approach is indeed viable, meaning all other models demonstrate good agreement as well. It was speculated, however, that unsteady effects might be more pronounced at higher frequencies. Regarding other numerical models, this time a fully resolved CFD model showed the most accurate results for both thrust and torque in steady and unsteady scenarios.

An extensive validation campaign of various numerical tools was carried out within the OC6 Project Phase III and reported in Bergua et al. (2022). A large number of participants submitted their simulation results from various methods and codes ranging from BEM theory with static aerofoil model to fully resolved CFD simulations. Overall, it was concluded that the agreement between different models and the experiment was good for the particular combinations of frequencies and amplitudes studied. However, it was again emphasised that, due to the decision of the UNAFLOW experiment to keep AoA on the blades within the linear region of aerofoil coefficients curves, the conditions might have been easier for the models to handle.

To compare the models in more unsteady conditions, Bergua et al. (2022) introduced

two more test cases, in which either rotor speed or blade pitch changed during a surge cycle. Supposedly, in these cases the blades are more likely to experience dynamic stall and rapid change of the local AoA, leading to more pronounced unsteady flow. Unfortunately, there was no experimental data for validation, and the models were compared against each other. The comparison of peak-to-peak thrust amplitude showed that higher-fidelity tools based on vortex methods and CFD were within 3% of each other, while BEM showed a discrepancy of up to 18%. These results might indicate the limitation of BEM of handling highly transient flows.

Further comparison of the various numerical tools and the experiment was carried out by Cioni et al. (2023) and focused on wake characteristics. Contrary to the conclusions regarding the integral rotor forces, which were consistent between different models and the experiment, wake measurements saw bigger discrepancies. Perhaps, this was the case because at the examined distance (up to 2.3D downwind a turbine), various blade discretisation approaches (vortex elements, ADM, ALM, blade-resolved) played a bigger role on field variables (velocity) than on integral variables (thrust and torque). Despite the discrepancies, it was observed that all numerical models tend to overpredict the wake recovery rate, albeit to a varying degree. No clear evidence was found to explain this behaviour, as the results varied from participant to participant even within the same family of methods. Perhaps, this can be used to highlight the challenges of taking accurate measurements in the experiment as well as in setting up numerical simulations correctly.

The data set of the UNAFLOW experiment is invaluable for investigating the performance of different models and this project will also make use of it for validation purposes. Overall, the available studies give the impression that available numerical models provide good results for unsteady wind turbine rotor characteristics. However, investigation of a wider range of operating conditions might be beneficial for evaluating the performance in more extreme conditions.

2.3 Other works

Despite the need to characterise the basic aerodynamic performance of a moving rotor, it does not suffice to study it only by itself, in isolation. Some authors are taking their investigations further, attempting to account for more phenomena of interest, for example, by including a structural response of blades, or by considering turbines in wind farm arrangements. Some examples of such studies are given below.

Kheirabadi and Nagamune (2021) report on their efforts to create a simulation tool

capable of capturing floating platform motion and time-varying magnitude and direction of the wind. To allow for computations being performed in reasonable time-frames, low fidelity models as well some approximations were made. The authors claim, however, that the capability of the code will be extended in the future.

Rezaeiha and Micallef (2021) studied two floating offshore wind turbines positioned in line one after another. The upstream turbine was subjected to prescribed surging motion, while the downstream turbine remained fixed. An Actuator Disk Model was employed to model the turbine rotors, while the flow was solved with unsteady Reynolds-averaged Navier-Stokes equations. Power performance of turbines and their wake interactions were investigated to inform the design of FOWT farms. The results suggest that the power and thrust coefficients fluctuate in response to surge motion. Their mean values, however, remain quite similar to those obtained in the case of stationary rotors. The wake of the surging turbine was found to recover more quickly, following from which authors suggest that there might be potential for a more compact arrangement of FOWTs.

When considering multiple phenomena, it seems that researchers prefer lower-fidelity methods to CFD, because they lend themselves to easier coupling of different physical models. Ramos-García, Kontos, et al. (2022) used a vortex-based solver to predict the performance of a FOWT and further enhanced the methodology by considering structural dynamics, control algorithms, and hydrodynamic response due to mooring. Despite the use of many additional models, the study focused on overall aerodynamic characteristics and the effects of each of the additional phenomena were not distinguished. The response in pitch motion only and in surge motion only were investigated separately. Evidently, pitch has a more pronounced impact on wake development than surge. In fact, it was found that, compared to pitch, surge had little effect on the wake recovery rate, which agrees with the observations of Cioni et al. (2023), while contradicting original expectations.

In their subsequent work, Ramos-García, González Horcas, et al. (2022) looked at wind turbine performance in wakes. It was shown that faster wake recovery of a pitching rotor allows to increase power output of downstream turbines. However, the gain tends to depend on the atmospheric turbulence intensity, which reduces the gain when increased. Furthermore, the oscillatory nature of turbine motion, and, consequently, of its wake, may induce resonant effects and lead to increase of downwind floater motions.

Similar to the previous study, Castorrini et al. (2023) combined CFD and finite-element analysis (FEA) to assess structural response of a floating wind turbine in pitch. The aerodynamics were modelled using ALM for blades and DES for the wake, while 1D FEA solver computed the structural response. The authors highlight that elastic response

of blades is amplified in case of a floating turbine and should be accounted for when estimating blade loads.

Other works do not explicitly mention floating wind turbines but are useful for offshore wind in general. For example, Sorensen et al. (2018) and Yan and Archer (2018) discussed the effects of compressibility on wind turbine aerodynamics. The results suggest that compressibility effects are negligible at low to optimal tip speed ratios (TSRs), while at high TSRs, the difference is more apparent. In terms of their effect on long-term operation, however, it is expected to be limited as turbines rarely operate at high TSRs.

Regarding the environmental conditions, K. S. Hansen et al. (2012) discussed the effect of atmospheric turbulence, whereas Dörenkämper et al. (2015) focused on the effect of an atmospheric boundary layer on wind farm performance. Control strategies also offer the potential for improving wind farm efficiency. Vollmer et al. (2016) studied the effect of intentional yaw misalignment on wake propagation, while Zalkind et al. (2022) suggested a framework for optimising a wind turbine controller.

Overall, as mentioned at the beginning of this section, offshore wind is a multidisciplinary area of research, and sometimes it is difficult to predict which phenomena might have the largest influence on the performance of the whole system. Therefore, studies which combine a full range of relevant physics models are valuable.

3 Methodology

The section gives an overview of techniques and procedures that were used to set up and run the simulations. It begins with an overview of governing equations used in CFD in sections 3.1 and 3.2. Next, relevant details about the experiment are described in section 3.3. Summary of mesh preparation and implementation of mesh movement is given in sections 3.4 and 3.5. It is then followed by details about the numerical setup in section 3.6. Finally, the definition of parameters which will be investigated is given in section 3.7, and some notes on the limitations of the study are discussed in section 3.8.

3.1 Governing equations of fluid flow

The flow fluid is governed by a set of conservation principles collectively known as the Navier-Stokes equations (NSE). The first equation represents the conservation of momentum principle and can be written in vector form as (Cengel and Cimbala, 2017)

$$\rho\frac{\partial\vec{U}}{\partial t} + \rho(\vec{U}\vec{\nabla})\vec{U} = \nabla p + \rho\vec{g} + \mu\nabla\vec{U} \tag{1}$$

Each term has its associated physical meaning, namely

- $\rho\frac{\partial\vec{U}}{\partial t}$ is the unsteady term, describing the rate of change of momentum with time;
- $\rho(\vec{U}\vec{\nabla})\vec{U}$ is the convective term, describing the rate of transport of momentum in space;
- $\mu\nabla\vec{U}$ is the diffusive term, describing the rate of change of momentum due to viscous effects;
- ∇p and $\rho\vec{g}$ are the source terms, describing the rate of change of momentum due to pressure gradient and gravity.

The second equation is the conservation of mass principle, which can be written in the form of the continuity equation as

$$\frac{\partial\rho}{\partial t} + \vec{\nabla}(\rho\vec{U}) = 0 \tag{2}$$

Similar to eq. 1, each term in eq. 2 accounts for a net rate of change of the quantity of interest, in this case specific mass, i.e., density. The first term depicts the rate of increase of mass with time, while the second one refers to the net rate of mass flow into or out of a fluid element (Versteeg and Malalasekera, 2007).

Additional transport equations for quantities of interest can be considered, such as energy, turbulence properties, or species. Turbulence modelling is addressed in the next section, while other quantities are neglected in this work.

Eq. 1 can be written in scalar form for each of the three velocity components. These, together with eq. 2, form a system of four equations with four unknowns (three velocity components and pressure). The NSE are unsteady, non-linear, second-order partial differential equations (PDE) and cannot be solved analytically for a general three-dimensional problem. Incidentally, the proof of existence and smoothness of the solution constitutes one of the seven Millennium Problems (Clay Mathematics Institue, 2023).

Some of the ways of solving the NSE require significant approximations, such as reducing the number of dimensions (one-dimensional axisymmetric flows), limiting the speed (creeping flows), or making other approximations (e.g., potential flow theory) (Ferziger and Peric, 2012). On the other hand, computational fluid dynamics (CFD) attempts to solve the complete equations by discretising them into a set of linear algebraic equations, for which solution algorithms exist.

A popular method for discretising the NS equations is known as the Finite Volume Method (FVM). Its main feature is the use of an integral form of the conservation equations as a starting point for discretisation (Ferziger and Peric, 2012). In principle, the method ensures the conservation principle by design and therefore is expected to give physically realistic results. However, as part of discretisation process, assumptions about the distribution of field parameters in space and time need to be made. The choice of discretisation schemes is crucial in ensuring the stability and accuracy of the solution process and is discussed in more detail in section 3.6.

Various software packages are available for CFD modelling. This project will make use of a CFD toolbox OpenFOAM version v2106 (OpenFOAM: User Guide, 2023), which is open source, i. e., free, does not have a licencing restriction on the number of parallel processes, and is widely used in the academic community. OpenFOAM provides a variety of solvers suitable for a wide range of CFD problems. For this project, an incompressible, transient, single-phase solver `pimpleFoam` will be used.

3.2 Turbulence

It was observed and initially documented by Reynolds (1895) that, in certain flow regimes, fluid properties experience random fluctuations about their mean values. This phenomenon is now known as turbulence and is present in flows with high Reynolds number,

when characteristic length and/or speed are sufficiently large, which is the case for most engineering flows. Turbulence plays an important role in energy transfer within the flow and therefore needs to be considered when solving such fluid dynamics problems. There is a wide choice of turbulence modelling approaches, and they can be broadly divided into two families, the ones based on Reynolds averaging and the Scale-Resolving Simulations.

In Reynolds averaging, an instantaneous flow property is expressed as a sum of its average time-independent value and a random time-dependent turbulent contribution (Versteeg and Malalasekera, 2007). Substitution of this decomposition back into NSE allows to separate the terms with only mean flow values from the terms with turbulent fluctuations. The resulting set of equations is then closed with additional relationships for the new quantities. The nature and number of these equations vary from one-equation algebraic models (mixing length model) to 6-equation PDE model (Reynolds stress equation model). The widely known $k-\epsilon$ and $k-\omega$ models belong to this family.

The Scale-Resolving Simulations attempt to solve the equations on sufficiently fine grids with sufficiently small time-steps to resolve the required length and time scales. The models include large eddy simulation (LES) and direct numerical simulation (DNS).

When it comes to modelling wind turbines, the two most prominent approaches seem to be the $k-\omega$ model and the LES model. For example, the works of Tran and Kim (2016), Tabib et al. (2017), and Cormier et al. (2018) employ the $k-\omega$ SST model, while Mancini et al. (2020), Strickland and Stevens (2020), and Maas and Raasch (2022) use the LES model. Since LES is more computationally expensive than RANS, it tends to be paired with lower-fidelity approaches to represent the blade geometry, such as the actuator disk model, or the actuator line method (Andersen, Sørensen, and Mikkelsen, 2013). Blade-resolved simulations, on the other hand, make use of the less demanding RANS model. For this work, the RANS $k-\omega$ model was chosen.

The $k-\omega$ model is a linear two-equation turbulence model . It uses the Boussinesq hypothesis of linear relationship between turbulent stress and strain rate with the constant of proportionality equal to turbulent eddy viscosity. The eddy viscosity is computed through the use of turbulence kinetic energy k and turbulence specific dissipation rate ω, each determined by the solution of a corresponding partial differential equation.

The shear stress transport (SST) formulation of the $k-\omega$ model was introduced by Menter, Kuntz, and Langtry (2003) and can be used for a wide range of engineering problems. A characteristic feature of the model is the use of a blending function for wall treatment. The model automatically switches between low-Re near-wall treatment to far-field high-Re treatment with wall functions, depending on the grid resolution near

the wall. Menter, Kuntz, and Langtry (2003) show that this approach allows to obtain accurate velocity profiles irrespective of y^+ values. As the authors highlight, this removes a constraint of $y^+ < 2$, which might be excessive for industrial applications such as the one in this project.

3.3 Experiment details

When conducting numerical experiments, it is desirable to have a benchmark to compare obtained results against. Usually considered benchmarks are either an analytical solution or experimental measurements. A floating offshore wind turbine is a complex engineering system, which is virtually impossible to completely describe analytically using only mathematical relationships. This leaves experimental data to be the only viable source of validation. As was discussed in section 2.2, the UNAFLOW project has freely available data on a surging wind turbine tested in a wind tunnel. The data was also used in the OC6 Phase III project (Robertson et al., 2023), where it was compared to the results from various numerical models. It was decided to make use of this available dataset.

Robertson et al. (2023) recommended to use three Load Cases 2.1, 2.5, and 2.7 for simulations using CFD because both forces and wake velocity profiles were measured for them. Due to time limitations, this project focused only on one of these test cases. Load Case 2.7 was chosen as it has the highest surge frequency, which presumably will allow to obtain a full cycle results faster than for other, lower, frequencies. Important experimental parameters are listed in table 1.

Additionally, a 3D CAD model of the blade was made available to the participants of the project and was used here as well. The full description of the blade geometry can be found in Robertson et al. (2023) with further details in Bayati, Belloli, Bernini, Giberti, et al. (2017) and Bayati, Belloli, Bernini, and Zasso (2017). The model was developed as a scaled version of the DTU 10 MW turbine. However, instead of purely geometric scaling, the blades were modified to produce the same thrust and torque coefficients as the full-

Table 1: Load Case 2.7 simulation parameters

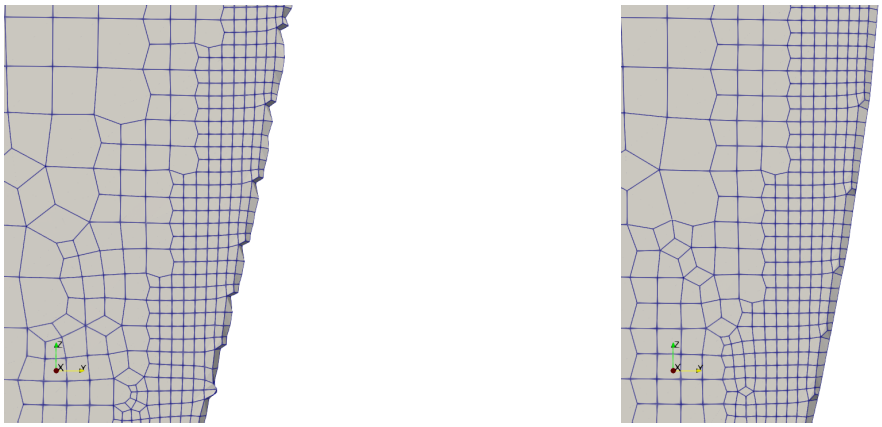
Parameter	Units	Value
Wind speed	m/s	4.19
Rotor speed	rpm (rad/s)	241 (25.13274)
Surge frequency	Hz (rad/s)	2.0 (12.56637)
Surge amplitude	m	0.008

scale prototype but at a smaller (model scale) Reynolds number. Further modifications to the geometry were made to facilitate a more efficient meshing process and are described below.

3.3.1 Modification of the blade geometry

The provided 3D model of the blade has a sharp trailing edge and tip. These infinitely small features prove to be difficult for the meshing utility to handle, which results in a big number of highly skewed faces and cells at the thinnest regions of the model, as shown in figure 1a. Refining the mesh at the blade edges does not help, as these irregularities keep reducing in size but cannot reliably fit to the TE. It is speculated that this constitutes a limitation of the meshing tool `cfMesh`, which creates a mesh by continuously refining an initial Cartesian hexahedral mesh. Since the curved geometry cannot be perfectly aligned with the mesh nodes, there are always a few cells within proximity of the curved feature. Due to the infinitely small size of said feature, the snapping process cannot reliably detect where to move the node to fit the curve, which ultimately results in a serrated pattern.

The resulting skewed cells may negatively affect the stability of computations and accuracy of final results. To address this potential issue early on, it was decided to round off the trailing edge in a similar way to the 15MW reference wind turbine developed within the IEA Wind TCP Task 37. Although the base design of the turbine also features a sharp TE and a tip reducing to a point, for CFD purposes the CAD model was modified to have a blunt trailing edge (IEA Wind Task 37, 2023). In this project, however, the TE was rounded, as it gave better results on a relatively coarse mesh (see figure 1b). The rounding was performed with the "Chamfer" tool in SOLIDWORKS, using the fillet feature type



(a) Sharp TE **(b)** Rounded TE

Figure 1: Comparison of surface meshes resulting from a blade with a sharp and rounded TE

with chord width equal to $0.5mm$ and a circular profile, i.e., the resulting TE thickness immediately before the rounding is $0.5mm$ and the curvature radius is approximately $0.25mm$. Figure 2a illustrates the comparison between the outline of the initial blade and the resulting one. Additionally, figures 2b and 2c depict some characteristic dimensions at the tip of the modified blade.

To further quantify the changes made, the surface area of the blades was estimated with the help of the "Evaluate - Mass Properties" tool in SOLIDWORKS. The results are given in table 2. The relative difference between the designs is 3.8%. Regarding the potential effect on the simulation results, it may be speculated that, considering the conventional formula for aerodynamic force, reduction in surface area may lead to proportional reduction in force, provided that the thrust and torque coefficients remain the same. However, the majority of the force generated on an aerofoil happens at around its thickest part (Anderson, 2017), and therefore the loss of area at the trailing edge will most likely have small effect. That said, the assumption of constant force coefficients seems to be a crude one, as the change in geometry will most likely result in the change of polars of the blade aerofoils and of the flow field behind it. The nature of these changes, however, is difficult to quantify. Furthermore, in Robertson et al. (2023) it was not reported to what accuracy the blade in the model experiment was manufactured, so it is difficult to estimate the effect of changes in relation to experimental results. Consequently, the effect of these changes may be regarded as an uncertainty, rather than an error.

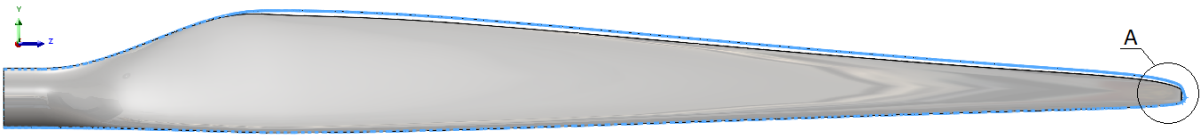
Finally, the modified blades were mounted on a hub, 3D CAD model for which was also provided, and pitched to 0 deg. Nacelle and tower were not included in the final geometry, similar to the majority of other project participants. Figure 3 gives an overview of the full rotor, while table 3 lists its characteristic parameters.

3.3.2 Modelling a turbine in a wake

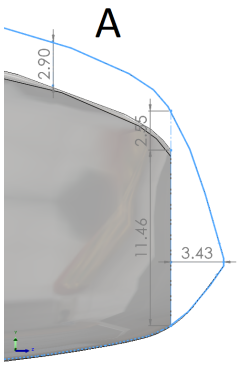
One of the aims of the project is to evaluate the performance of a rotor in a wake. Modelling two rotors explicitly might require computational and time resources not available

Table 2: Comparison of surface area of the original and modified blades

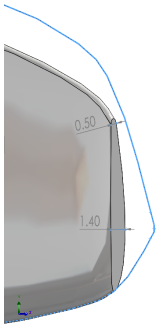
Parameter	Value
Surface area of the original blade	$0.1763m^2$
Surface area of the modified blade	$0.1696m^2$
Difference	3.8%



(a) Plan view of the full blade



(b) Close up at the tip (dimensions in mm)



(c) Isoview of the tip (dimensions in mm)

Figure 2: Comparison of contours of the initial blade with a sharp TE (outlined in blue) and the modified blade with a rounded TE (shaded in grey)



Figure 3: View of the rotor model used for the simulations

Table 3: Main characteristics of the rotor

Parameter	Units	Initial design (Robertson et al., 2023)	Modified design
Rotor diameter	mm	2381	2370
Blade length	mm	1102	1096
Start of rounded TE (from blade root)	mm	-	170
Hub diameter	mm	178	178
Blade pitch angle	deg	0	0
Rotor swept area	m^2	4.45	4.41

for this project. As a result, the wake is introduced by using a so-called cyclic boundary condition at the outlet and the inlet of the computational domain. This condition makes both inlet and outlet share field variables, which means that the wake created by a turbine within the domain will be mapped back on the inlet boundary upon reaching the outlet. In essence, such arrangement represents a turbine in an infinite array of turbines.

The expected evolution of the solution is as follows. The solution is initialised with a constant wind speed throughout the domain equal to the far-field wind speed. Initially, the turbine can be considered as standing in the first row of an imaginary wind farm. After a time period equal to domain length over wind speed, the wake will reach the turbine, and it can be now considered as representing a second-row turbine. In principle, this process will continue repeating itself infinitely. However, the simulation will be run for only a time-period available for this project.

3.4 Mesh generation

The computational domain has a cylindrical shape owing to overall axial symmetry of the problem (see figure 4). It has a round cross section with diameter roughly 5D, similar to other studies such as Rasam et al. (2016), Tran and Kim (2016), and Tabib et al. (2017). This diameter corresponds to wind turbine blockage, expressed as a ratio of rotor swept area to domain cross-section, of

$$\frac{\pi R_r^2}{\pi R_d^2} = \frac{1.185^2}{6.0^2} \approx 0.039, \tag{3}$$

where R_r is the rotor radius, m; R_d is the domain cross-section radius, m. The resulting blockage ratio is 3.9%.

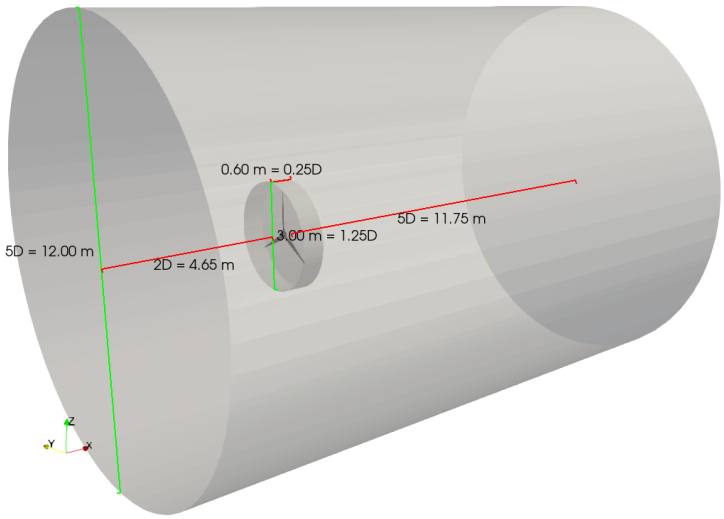


Figure 4: Main dimensions of the computational domain

One of the aims of this project is to simulate a wind turbine in a wake. Due to the decision to use the cyclic boundary condition on the inlet and outlet boundaries, the total length of the domain should be equal to the spacing between (imaginary) wind turbines. Consequently, the total length of the domain was chosen to be equal to 7D, which is an average value of spacings found in practice (Archer et al., 2018).

The same domain will be used to model isolated wind turbine and a rotor in its own wake. As a result, the inlet boundary is placed 2D upstream and the outlet boundary is placed 5D downstream. This is in contrast to placing the rotor in the middle of the domain, which, perhaps, would make more sense from the point of view of distancing the rotor father from the boundaries for a purely cyclic case.

The computational grid was generated using a `cfMesh` utility supplied with OpenFOAM. Table A1 of Appendix A gives an overview of the main parameters used in `meshDict`, a dictionary with mesh generation controls. Where possible, the refinement regions were defined with the `refinementLevel` setting. This way, the actual cell size depends only on the `maxCellSize` setting, which makes changing it for the mesh convergence study straightforward and systematic. Figures 5 and 6 give an overview of the resulting mesh.

Some of the expected important features of the flow around the rotor are a blade-tip and a blade-root vortices. To better resolve them, additional refinement regions were specified to cover their expected paths. In a test simulation it was observed that a cylindrical refinement region with a round cross-section of diameter 0.4m covers the root vortex, while a hollow cylindrical refinement region with an inner diameter of 1.0m and an outer diameter of 1.5m covers the tip vortex.

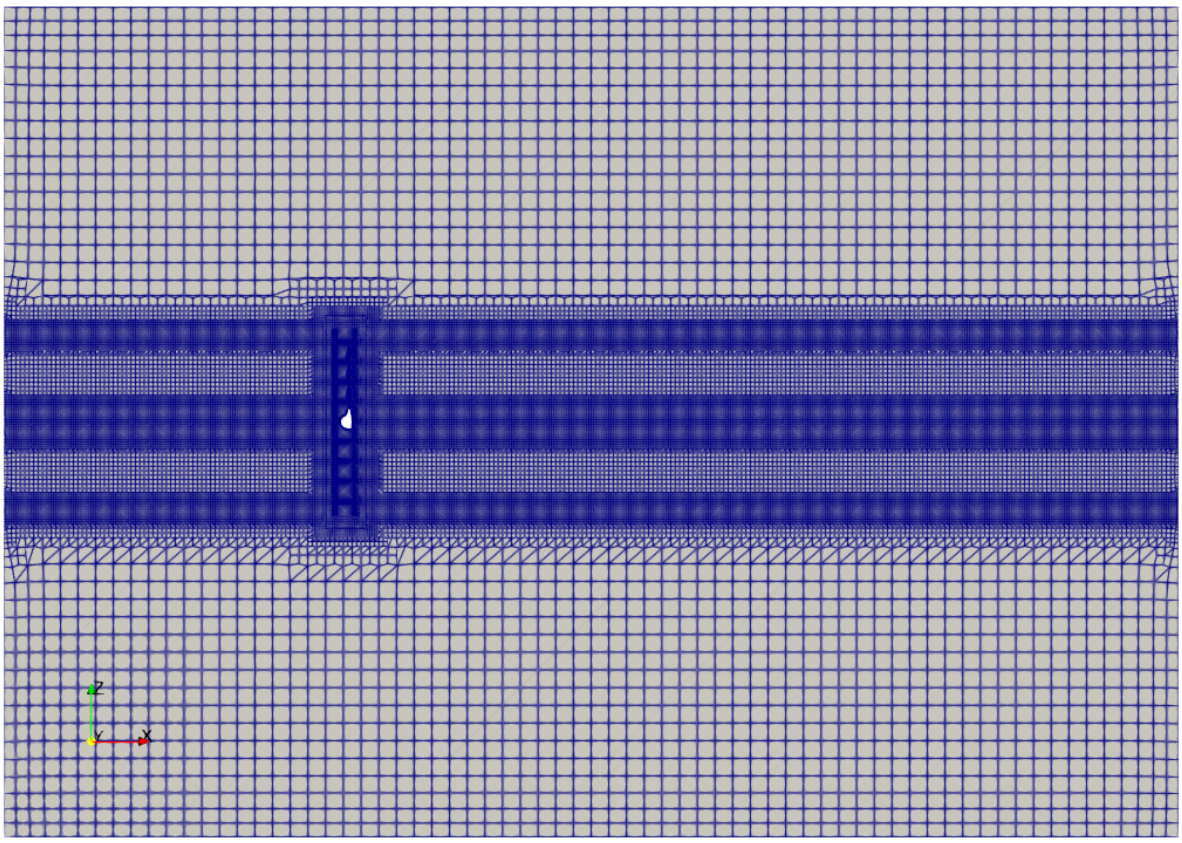
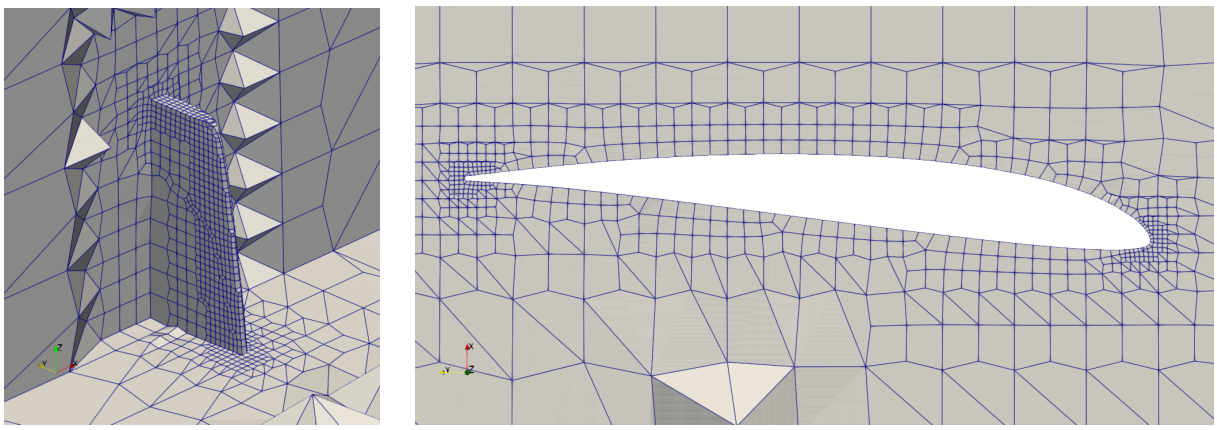


Figure 5: Computational mesh (a cross section along the centerline of the domain)



(a) Mesh around the tip

(b) Mesh around blade section at r=0.7m

Figure 6: Computational mesh (a closer view)

Regarding the rotor itself, additional refinement levels were specified at the leading edge, trailing edge, and tip to better capture small curvatures. In an attempt to reduce the mesh cell count to limit computational costs, the mesh was kept relatively coarse and the prismatic layers at the walls were not added.

It is also worth noting that the `cfMesh` utility does not seem to allow to control the number of cells between different refinement levels the way that the `snappyHexMesh` utility does with the `nCellsBetweenLevels` setting. As a result, the mesh has areas where the cell size changes rapidly by a factor of 2 multiple times. This is not a good practice as the rapid change in mesh size may be detrimental to the accuracy of the results. Manually creating additional 3D regions for refinement was rather cumbersome and inflexible and created refinements where it was not necessary. The final mesh was left as is, but the limitation was noted for further investigation in future works.

3.5 Mesh motion

To model rotor rotation and surge, two approaches have been investigated. The first one uses the overset meshes technique, while the second one combines sliding mesh for rotation and morphing mesh for surge.

3.5.1 Overset meshes

Overset meshes technique is considered to be efficient at handling complex geometries (Khaware, Gupta, and Rajan, 2018). It allows to mesh the solid body(ies) separately from the domain, which can facilitate the use of structured meshes for each of the component meshes, which might be impossible if all bodies are to be meshed together. Furthermore, it allows to resolve movement with large amplitudes without the loss of mesh quality. The disadvantages of the method are the possibilities of errors due to interpolation between different meshes, associated with interpolation additional computational effort and further additional computational expenses needed to recompute the stencil at every time step in case of mesh motion. Nevertheless, Chen et al. (2019) have shown that the overset meshes technique can reliably handle various hydrodynamic problems, while Cormier et al. (2018) used it for the simulation of a wind turbine.

The overset meshes method starts by generating completely separate component meshes, one for the region in the vicinity of the moving body (called foreground hereafter) and one for the surrounding domain (called background hereafter). An example of the component meshes is given in figures 7a and 7b. The background and foreground are merged

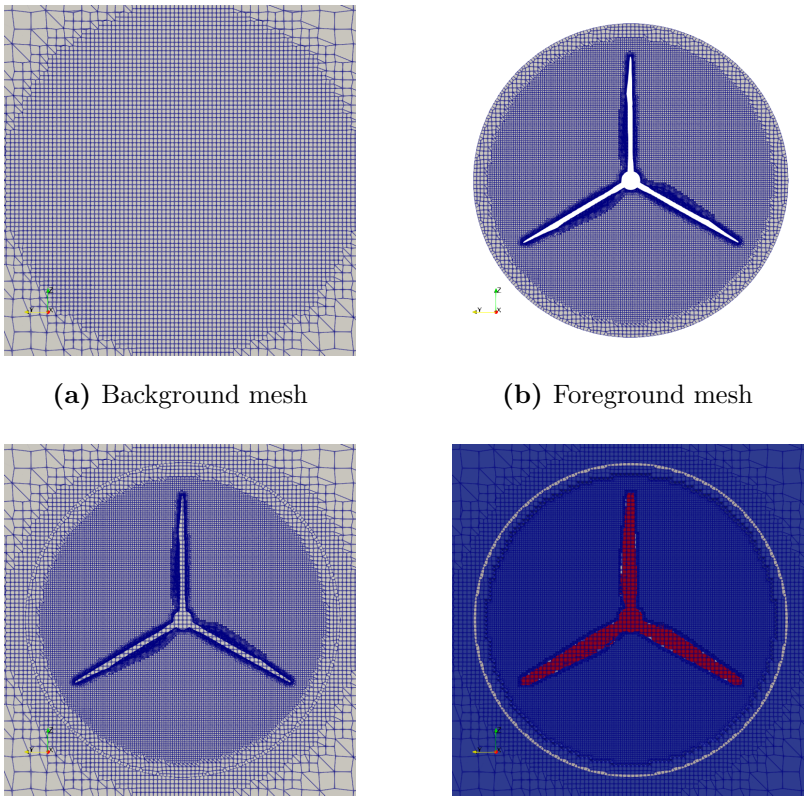
together to form a computational grid, however the regions remain topologically disconnected (see figure 7c). The next steps are handled by the solver and involve determining positions of the component meshes relative to each other in order to designate cells as "calculated", "interpolated", or "hidden". Figure 7d depicts the result of such operation, where red cells were identified as "hidden", grey cells as "interpolated", and blue cells as "calculated". As their names suggest, the solution is being computed in blue cells and interpolated between regions in grey cells, while red cells are excluded.

3.5.2 Sliding and morphing mesh

While overset meshes can model virtually any combination of motions in all of the 6 degrees of freedom, other approaches can be combined to account for both rotation and surge. Rotation can be included in the solution by using the sliding mesh technique. The sliding mesh is similar to the overset mesh in that two separate meshes are generated and then combined to form the final computational grid. Here, however, interpolation happens not in the fringe region of the foreground mesh, but on the defined boundary between the regions. This boundary is referred to as an arbitrary mesh interface (AMI). The AMI patches are physically touching and sliding against each other. This geometrical connection of the patches simplifies the interpolation process when compared to the one required for the overset meshes. Sliding mesh interface was shown to effectively handle rotating geometries (Tabib et al., 2017).

The surging motion can be modelled using the morphing mesh technique. The method can be successfully used for studying morphing aerofoils (Abdessemed, Yao, and Bouferrouk, 2021), since the deflections tend to be relatively small. However, it becomes more challenging to implement the method for larger motion amplitudes. For example, in their attempt to study a wave energy converter response amplitude operator, Katsidoniotaki and Göteman (2022) were unable to complete the simulations for the largest amplitudes, as the mesh deformation reduced its quality significantly, leading to divergence and a crash. However, they did observe that the solution was quicker than the one with the overset meshes technique.

Adding surge motion into the proposed case requires two steps. Firstly, the rotating region with the rotor is forced to move as a solid body. Secondly, the AMI interface of the stationary region must oscillate in the same manner in order to retain the physical (geometrical) connection with the AMI of the rotating region. To handle this motion, the morphing mesh motion solver `displacementLaplacian` is used. This solver treats



(a) Background mesh **(b)** Foreground mesh

(c) Combined mesh **(d)** Solution mesh

Figure 7: Assembly of an overset mesh

the displacement of mesh points as a field variable and solves a Laplace's equation for it. Given appropriate boundary conditions (i.e., motion laws of the boundaries), the obtained solution is diffused (propagated) inside the domain, modifying the location of surrounding points. Figure 8 illustrates the mesh morphing at different points in time.

The effect of cell deformation is expressed as a `meshPhi` variable, which stores the change of momentum flux due to the change in cell volume. It is then used in the equations for the actual field variables (e.g., velocity, pressure) to account for the mesh deformation.

The propagation of point displacement inside the domain can be adjusted by the diffusivity setting. The `exponential 2 inverseDistance` option was used. The `inverseDistance` part of the setting ensures that the mesh deformation increases with distance from the AMI boundary. This will allow to move deformation and associated numerical inaccuracies further away from the AMI boundaries. The exponential part of the setting determines the speed of this increase. Through trial and error, it was found that the `exponential 2` setting gave the best quality meshes within the used movement range in terms of keeping cell skewness low.

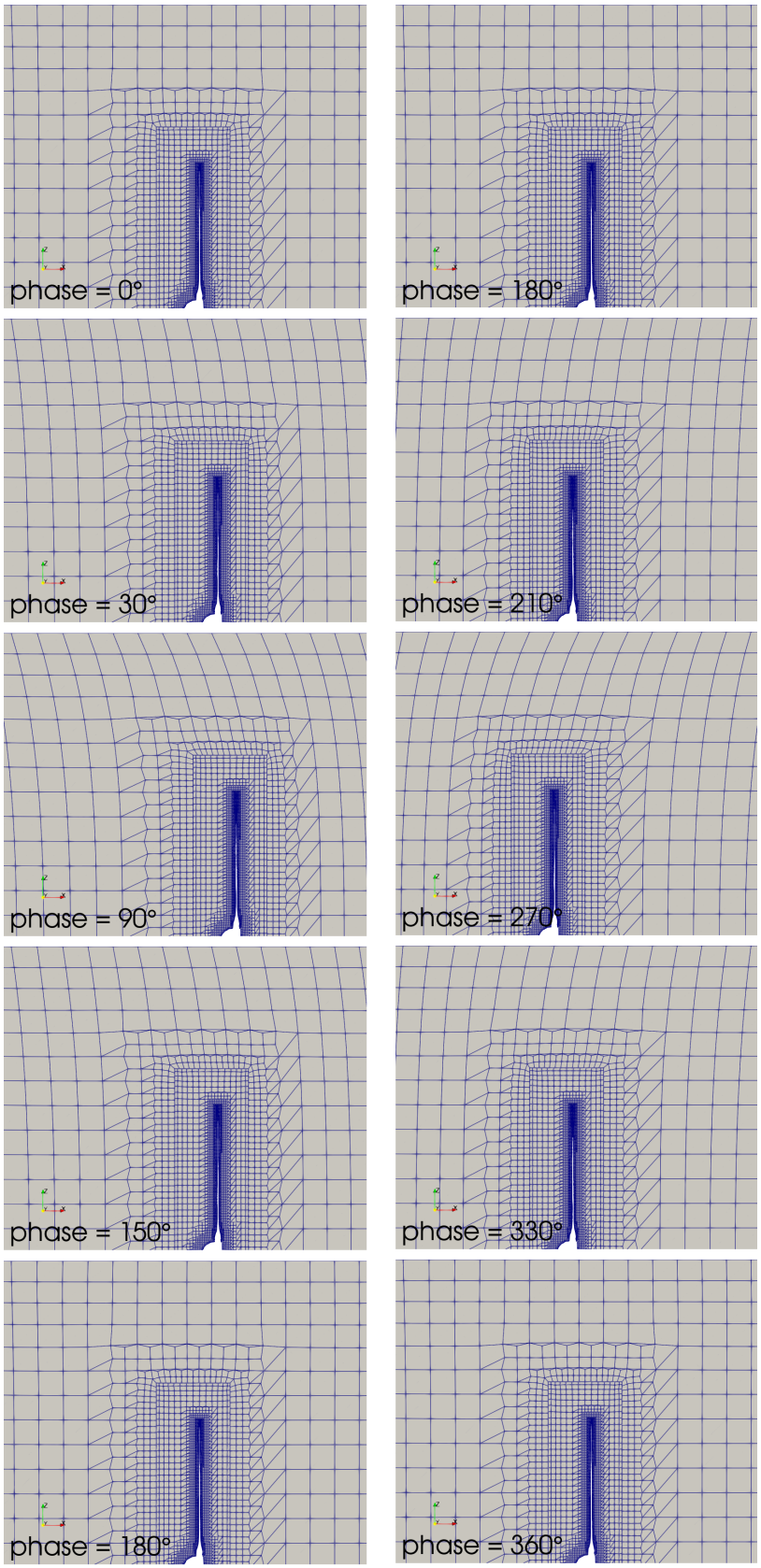


Figure 8: Illustration of a morphing mesh at different phases of a surge cycle (rotation disabled)

3.6 Numerical setup

3.6.1 Discretisation schemes

As part of a solution process, CFD requires the initial differential equations to be discretised in time and space to obtain a system of linear algebraic equations. The process requires a number of assumptions about a distribution function of field variables between grid nodes. The discretisation of each term can be performed with various methods. Generally speaking, a choice of a discretisation scheme is a compromise between its accuracy and stability. A trial-and-error approach was used to determine optimal settings. Initially, the most accurate schemes were selected and then downgraded to achieve a stably converging solution. In the end, the following discretisation schemes were used.

Time scheme The Euler scheme was selected for the discretisation of the unsteady term. The scheme is first-order accurate. It must be noted, however, that this setting only affects the first unsteady term in eq. 1. It appears that in OpenFOAM all other time dependent variables in other terms are always computed implicitly. An implicit scheme is unconditionally stable (Versteeg and Malalasekera, 2007) and therefore may tolerate relatively large time-steps, which is a desirable feature in this particular simulation.

The nature of the meshing technique used leads to the presence of both very large cells (0.3m in the far field) as well as very small cells (0.0005mm at the leading and trailing edges of the blade). These small cells are few in number but lead to a very small time-step, if the Courant criterion Co is to be satisfied. The use of implicit scheme may allow the use of a bigger time step with $Co > 1$ without a loss in stability and accuracy. To assess the viability of this approach, a time-step-dependence study will be carried out.

Gradient scheme OpenFOAM offers two gradient schemes, one based on the Gauss theorem with linear interpolation and the least squares one. The former is regarded as the first-order accurate scheme, while the latter is second-order accurate. However, Syrakos et al. (2017) report that, in locally distorted and arbitrary meshes (which is the case for the present work), the accuracy of the schemes reduces to zeroth- and first-order respectively. Nevertheless, the least-squares method remained more accurate than the Gauss-based method, and therefore was the preferred first choice.

Divergence scheme Discretisation of the convective term is challenging due to the need to account for flow direction, i.e., the scheme is required to have a property of

transportiveness (Versteeg and Malalasekera, 2007). As a result, a number of schemes have been developed and are available in OpenFOAM. Generally speaking, the linear scheme is the most accurate one. However, it is sensitive to the mesh resolution and gives unbounded results on coarse meshes, which may lead to solution divergence. On the other side of the spectrum is the upwind scheme, which is unconditionally bounded, but introduces high numerical diffusion. Other schemes tend to fall in between these two and offer a balance between accuracy and stability and are often referred to as total variance diminishing (TVD) schemes.

It appears that the choice of a divergence scheme depends on each simulation setup. For example, Tran and Kim (2016) used the upwind scheme. Tabib et al. (2017), on the other hand, used the linear scheme, while Andersen, Sørensen, and Mikkelsen (2013) employed a QUICK scheme. In a numerical setup used in this work, neither linear, nor other TVD schemes such as QUICK and van Leer gave stable results. The linear upwind scheme seemed to be stable and therefore was used throughout the simulations.

Other schemes OpenFOAM additionally requires the specification of schemes for all other terms. For these, the recommended options were used. A complete listing of the `fvSchemes` dictionary is given in Appendix B.1.

3.6.2 Solvers

The selection of solvers was left to the default recommended parameters. For pressure, a generalised Geometric-Algebraic Multigrid (GAMG) solver with a `DICGaussSeidel` smoother was used. The multigrid acceleration technique is used to increase the convergence rate by solving the equations for error on progressively coarsened grids. The actual solution is achieved with the combination of DIC and Gauss-Seidel iterative methods.

The overset meshes technique implemented in OpenFOAM does not support the GAMG solver. Therefore, the pressure solver was changed to `PCG` with a `DIC` preconditioner.

All other equations are solved using a `symmGaussSeidel` solver, which uses a symmetrical Gauss-Seidel iterative method. In this context, "symmetrical" refers to the use of forward and backward sweeps during the solution process, which increases the rate of propagation of the solution into the domain. Further details about the mathematical formulation of the solution algorithms can be found in Ferziger and Peric (2012). A complete listing of the `fvSolution` dictionary is given in Appendix B.2.

3.6.3 Initial and boundary conditions

The used boundary conditions correspond to the usual ones used in external fluid flow problems. The free stream velocity was defined at the inlet, while the reference pressure of 0 was set at the outlet. The sides of the domain were treated as slip walls. The rotor wall assumed the no-slip condition. Full specification of used boundary and initial conditions can be found in Appendix C.

One of the simulated test cases used a cyclic boundary condition at the inlet and outlet walls. In this arrangement, neither velocity nor reference pressure can be defined, and the case reduces to something akin to a "fan in a room", when the airflow due to wind is not accounted for. To address this, a `momentumSource` was added in the `fvOptions` dictionary. A `meanVelocityForce` setting accepts the desired mean velocity field (4.19 m/s in stream-wise direction) and applies the equivalent pressure gradient along the domain, which sustains the overall flow. A similar approach was used by Andersen, Sørensen, and Mikkelsen (2013).

OpenFOAM features the 2003 version of the $k - \omega SST$ model (OpenFOAM: User Guide, 2023). Initial conditions for the turbulence kinetic energy k and turbulence specific dissipation rate ω were estimated with the following formulae, suggested by the OpenFOAM user manual (OpenFOAM: User Guide, 2023).

$$k = \frac{3}{2}(Iu_{ref})^2 \tag{4}$$

where $I = 2\%$ is the turbulence intensity as estimated in the experimental campaign (Robertson et al., 2023) and $u_{ref} = 4.19\ m/s$ is the reference inlet velocity.

$$\omega = \frac{k^{0.5}}{C_\mu^{0.25} L} \tag{5}$$

where $C_\mu = 0.09$ is the turbulence model constant (OpenFOAM: User Guide, 2023) and L, m is the characteristic length scale assumed to be equal to the cell size at the inlet.

At the rotor wall, k is assumed to be 0, i.e., no turbulence, while ω was estimated using eq. 5 but with the characteristic length scale equal to the cell size closest to the wall.

These boundary and initial conditions were specified at the inlet boundary and in the internal field. The turbulence model requires the velocity at the first cell wall, which is not yet available at the solution iteration. As a result, wall functions are used at the wall to estimate the velocity in accordance with the law of the wall. Full specification of used boundary and initial conditions can be found in Appendix C.

3.7 Parameters

The following quantities will be monitored and compared:

- rotor-integral aerodynamic forces (thrust and torque), and
- axial velocity profiles in the wake in along-wind and cross-wind directions.

Following the best practice of comparing results, non-dimensional parameters will be used, where applicable. The normalisation is performed as follows.

- Thrust force is expressed as thrust coefficient C_t.
- Torque is expressed in terms of power coefficient C_p.
- Linear dimensions are normalised with respect to the rotor diameter $D = 2R = 2.37\ m$.
- Velocities are normalised with respect to the far-field wind speed $V = 4.19\ ms^{-1}$.

Cross-wake velocity profiles are also characterised by an average velocity in the wake. All mathematical formulae for the mentioned parameters are given in Appendix D.

Additionally, a discretisation error will be estimated using a procedure outlined in Appendix E.1.

3.8 Limitations

It is worth noting that the suggested study does not account for all possible aerodynamic phenomena experienced by a real turbine. Furthermore, the study is carried out for a scaled model size. Consequently, some simplifications were made and are briefly mentioned below.

Compressibility effects are not considered in this study. Due to model of the turbine being a scaled one, the tip trails at a speed of $v = 2\pi nR = 2\pi \cdot 4 \cdot 1.185 \approx 29.9 m/s$, which corresponds to the Mach number $Ma = \frac{v}{c} = \frac{29.9}{330} \approx 0.09$, which is below the threshold of 0.3 when compressibility starts to play a significant role. When it comes to a full-scale rotor, however, it has been shown that including compressibility effects might affect the accuracy of the results (Sorensen et al., 2018; Yan and Archer, 2018).

Another approximation of the study is the lack of atmospheric boundary layer. Similar to compressibility effects, it has an impact on full-size turbines and farms (K. S. Hansen

et al., 2012; Dörenkämper et al., 2015), but in the model scale testing used in this study it is not accounted for, which is consistent with the experiment.

Furthermore, the most severe state is simulated, when a turbine is located directly in the wake. In real-life situations such conditions are rare due to constant changes in wind direction and due to wind farm layout accounting for predominant wind direction. The effects of such misalignment were studied by, for example, Vollmer et al. (2016).

4 Results and discussion

This section presents the results obtained in the study. It begins with the verification of the numerical setup by examining mesh convergence, time-step convergence, and some other relevant test cases in section 4.1. The results are then validated against available data from the UNAFLOW experiment and OC6 Phase III participants in section 4.2. Finally, results for a turbine in a wake are given in section 4.3.

4.1 Convergence studies

A number of preliminary studies were carried out to determine the optimal simulation parameters and test some settings. These include mesh convergence, time-step convergence, relative performance of two sets of discretisation schemes, and verification of the implementation of the cyclic boundary condition.

4.1.1 Mesh convergence

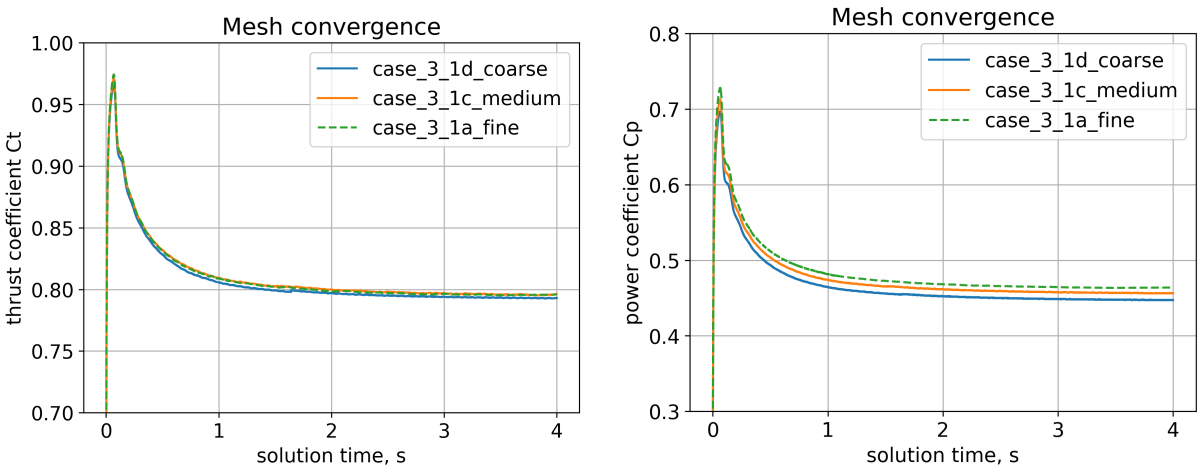
Firstly, a mesh convergence study was carried out, where results for three meshes were compared. The meshes were designated as fine, medium, and coarse. Table 4 gives an overview of characteristic features of the meshes used. It is worth noting that even the "fine" mesh is relatively coarse when compared to other studies, such as Cormier et al. (2018). However, for the purposes of this project it was deemed more important to obtain the results within the available time-frame rather than to attempt to reach maximum accuracy.

Figure 9 shows thrust and torque convergence histories on the three chosen meshes, while table 5 summarises the final results. The relative errors suggest that torque readings are more sensitive to the mesh refinement than thrust readings. Although the difference between the meshes still exists, it was decided to proceed with the finest mesh.

Based on the obtained results, a discretisation error was estimated in Appendix E.2.

Table 4: Meshes used for grid convergence study

Mesh	Relative cell size	Min cell size at the tip, m	Cell size in the wake, m	Max cell size in the far-field, m	Cell count
fine	1.00	0.00050	0.03200	0.25600	5,121,219
medium	1.20	0.00060	0.03840	0.30720	3,226,466
coarse	1.44	0.00072	0.04608	0.36864	2,085,311



(a) thrust **(b)** torque

Figure 9: Grid convergence results as illustrated by thrust and torque convergence histories

Table 5: Grid convergence study results

Mesh	Thrust coefficient		Power coefficient	
	value, -	difference, %	value, -	difference, %
fine	0.796	-	0.464	-
medium	0.796	-0.02	0.457	-1.59
coarse	0.793	-0.35	0.447	-3.53

4.1.2 Time-step convergence

Considering the time constraint of the project, it was deemed necessary to speed up the simulations. The time-step convergence study was carried out in an attempt to estimate the influence of large time-steps on key parameters. The time-steps ranged from $2.5e-6$ to $2e-5$ s. For the smallest time-step, the Courant number was reaching 0.6, while for the largest time step, the maximum Courant number reported by the solver reached 20. Increasing the Courant criterion above unity might be interpreted as that information travels faster than is captured by the solver, which might lead to the loss of flow features.

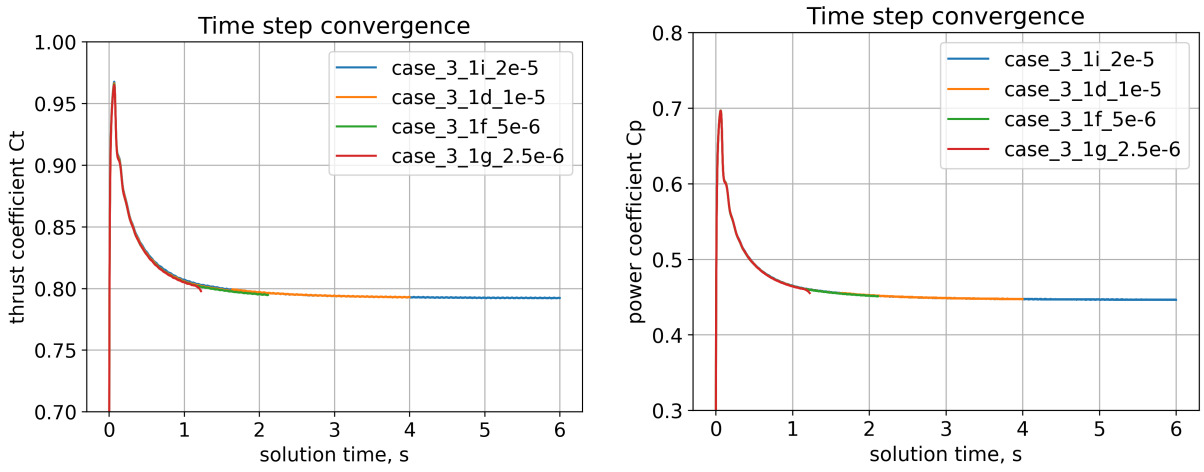
The use of the small time-steps proved to be extremely time consuming. As a result, final convergence of the readings was not reached. Nevertheless, figure 10 shows the time histories of thrust and torque measurements. The graphs seem to follow each other closely, which might suggest that, for that particular mesh, the chosen time steps produce essentially the same results as far as rotor integral forces are concerned. It is acknowledged that such conclusion cannot be definitive until the results have fully converged and recognised as a limitation of this study.

As for field variables such as pressure and velocity, figure 11 shows pressure and velocity readings at a point x=3.5m y=0.9m z=0m starting from solution time 1s. The shown data suggests that lower time-steps capture more fluctuations, however, the average values seem to agree well. It is assumed that using a large time-step is sufficient for this project.

Further investigation revealed that the cells where Courant number is larger than unity are limited to a small number located closer to the tip (see figure 12). Although this might indicate that the flow very close to the tip might not be sufficiently resolved, it does not seem to affect the thrust and torque on the rotor. For the purposes of this project, it was deemed that significant decrease in computation time far outweighs the potential error. Subsequent simulations will be run at time-step $2e-5$ s, which corresponds to $\approx 0.03deg$ of rotor rotation per time-step.

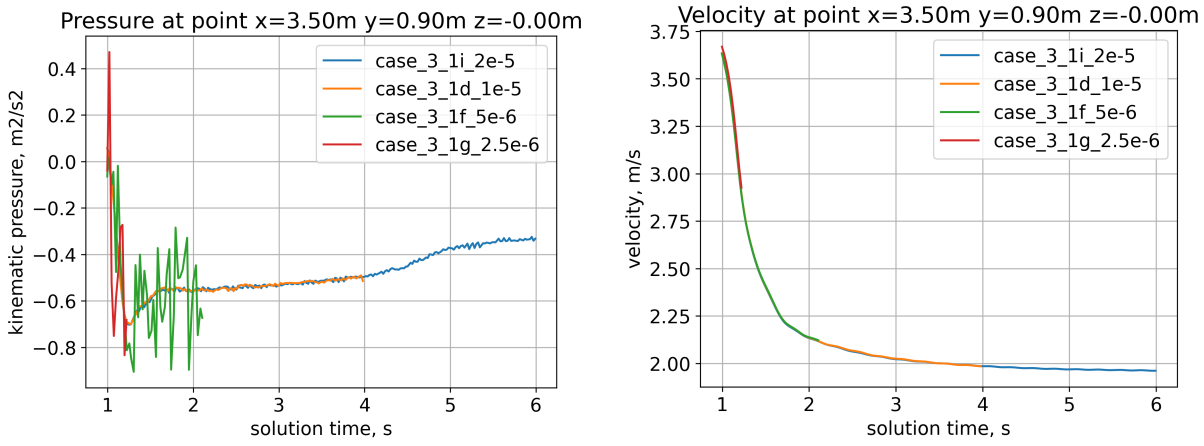
4.1.3 Other test cases

Comparison of discretisation schemes Two sets of discretisation schemes were tested. The so-called 1st-order schemes consisted of a linear scheme for gradient terms and an upwind scheme for divergence terms, while 2nd-order schemes combined a least squares approximation for gradient terms and a linear upwind scheme for divergence terms. Other settings were kept the same. It must be noted that the coarse mesh (see table 4) was used for this comparison. Figure 13 and table 6 give the obtained results.



(a) Thrust **(b)** Torque

Figure 10: Time-step convergence results as illustrated by thrust and torque convergence histories

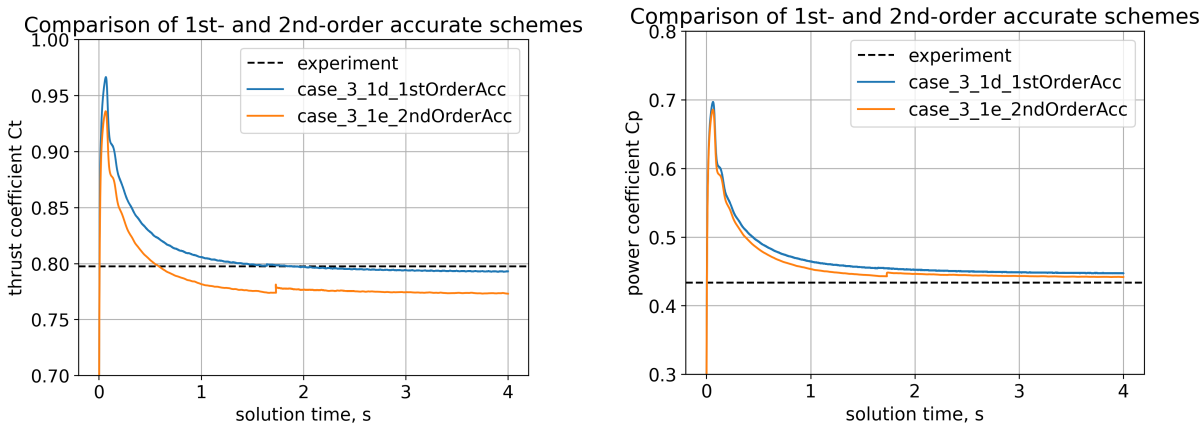


(a) pressure **(b)** velocity

Figure 11: Time-step convergence results as illustrated by pressure and velocity convergence histories



Figure 12: Cells where Courant number exceeds unity are depicted in red



(a) thrust **(b)** torque

Figure 13: Discretisation scheme convergence results as illustrated by thrust and torque convergence histories

Table 6: Discretisation scheme convergence results

Scheme set	Thrust coefficient		Power coefficient	
	value, -	difference, %	value, -	difference, %
experiment	0.798	-	0.433	-
1st-order accurate	0.793	-0.56	0.447	3.24
2nd-order accurate	0.773	-3.01	0.442	1.98

The results of different schemes may appear somewhat confusing, as the error of the 2nd-order accurate schemes increases for thrust coefficient but decreases for the power coefficient. This behaviour might be affected by the use of the coarse mesh for this comparison. As evident from the grid convergence study (see figure 9 and table 5), thrust tends to increase as the mesh is refined. The use of higher-order schemes, on the other hand, decreases the value. Such contradictory behaviour might be affecting an asymptotical convergence to the "correct" value. In other words, assuming that the results of both schemes on the finer mesh will increase by the same number of percentage points (say, 2% using error estimation parameters from table E1), the 2nd-order schemes will then give a result closer to the experimental value, confirming that higher-order schemes produce more accurate results.

As for the torque coefficient, both mesh refinement and higher-order schemes decrease the value. Such behaviour allows to come to a conclusion that both parameters asymptotically converge to the "correct" value.

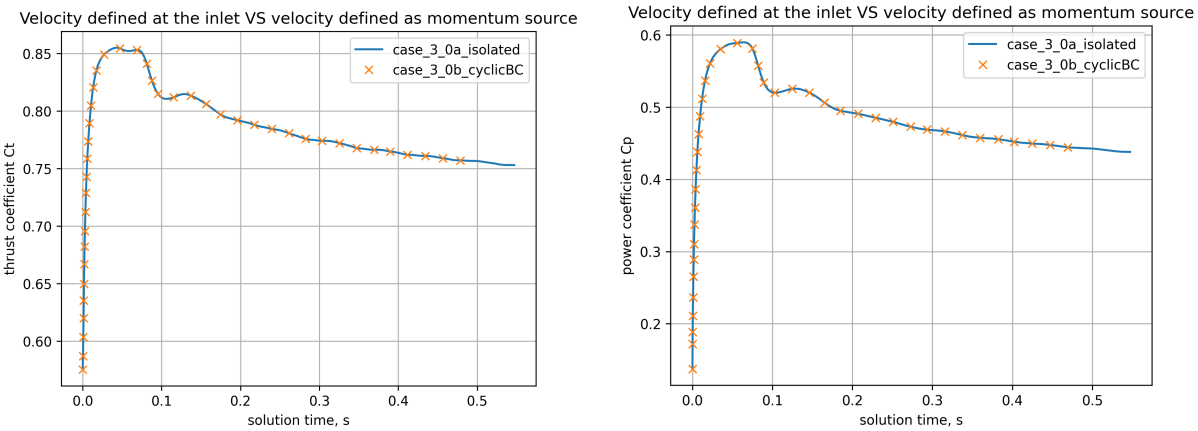
For the subsequent simulations, the 2nd-order schemes were used.

Limitation of the overset meshes technique During test simulations, it was observed that the overset meshes technique was considerably slower than the technique based on sliding mesh interface and mesh morphing. Perhaps, higher efficiency can be achieved with more careful numerical setup. It was deemed, however, to be outside the scope of this project, and all simulations reported here were performed using the second method.

Verification of cyclic boundary condition One more verification was carried out which investigated the behaviour of a test case with cyclic boundary conditions. Bearing in mind that the case is transient, it is expected that the influence of the cyclic boundary condition will not be evident at the beginning of the simulation, before the flow propagates far enough. Figure 14 demonstrates that, initially, before the wake reaches the turbine, the solutions in both cases are essentially the same. As such, it is assumed that the setting works as intended.

4.2 Comparison with the experiment

Next, results of simulations with chosen settings ("fine" mesh, $2E-5$ s time-step, 2nd-order accurate schemes) are compared with the experimental data from the UNAFLOW campaign. Please note that "steady" and "fixed" refer to a case when only rotor rotation



(a) Thrust **(b)** Torque

Figure 14: Comparison of thrust and torque between inlet and momentum source velocity definition cases

is considered, while "unsteady" or "surging" refer to a case when a rotor both rotates and surges.

Each simulation was run on one compute node of ARCHER2 cluster with 128 cores. 1 s of solution time took approximately 5 days to complete. Perhaps more careful choice of some settings, such as decomposition methods or solvers with higher parallel performance, could reduce the run-time, but that would require additional time to configure, which was not available for this project.

4.2.1 Steady rotation

Firstly, the rotor integral forces on a steady turbine are compared in table 7. The relative error to the experiment is -1.6% for thrust and 7.5% for torque. Lower accuracy of torque measurement is somewhat expected, as in the mesh convergence study it was observed that torque has not fully converged (see figure 9 and table 5). Further mesh refinement will likely allow to achieve better agreement. Incidentally, the observed error with respect to the experiment is quite close to the error estimated by the analysis of mesh convergence (6.6%, see table E1).

A number of CFD participants of the OC6 Phase III comparative study (Bergua et al., 2022) also reported that the accuracy of thrust prediction is higher than the accuracy of torque prediction, and there were studies which both overpredicted as well as underpredicted the torque value by $\pm 5\%$.

Next, velocity profiles in the wake are compared. Figure 15a shows along-wake velocities, while figure 15b depicts a cross-wake velocity profile. Additionally, results of other

Table 7: Comparison with the experiment (fixed isolated rotor)

Results of	Thrust coefficient		Power coefficient	
	value, -	difference, %	value, -	difference, %
experiment	0.798	-	0.433	-
CFD	0.785	-1.59	0.466	7.57

CFD participants from the OC6 study are shown as dotted lines. Along-wake velocities tend to be overpredicted closer to the rotor and slightly underpredicted further downstream when compared to the experimental results. However, the RMS value of the error is quite low at just 0.02 (4.8%), or 0.09 ms^{-1} in absolute terms.

Regarding the cross-wake velocity profiles, the RMS value of the error is larger at 0.11 (15.9%), or 0.48 ms^{-1}. Furthermore, some discrepancies might be observed. Firstly, the wake directly behind the hub is clearly overpredicted, which is consistent with most other CFD studies which did not include hub, nacelle, and tower geometries. Only CENER modelled all these and obtained lower wind speeds.

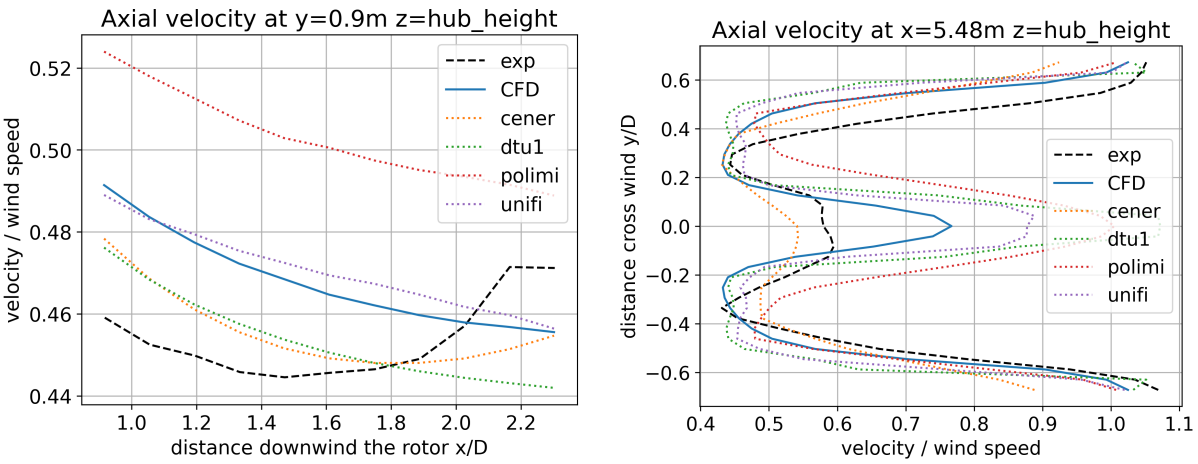
Moreover, it appears that experimental velocity profile has a slight axial asymmetry, while numerical results seem to be symmetrical. One more region where discrepancy is high is around the tip vortex. CFD results tend to show wider areas with low speeds. It is speculated that this happened due to wind tunnel blockage and proximity of the walls, which constrained wake expansion during the experiment. Narrower wake also resulted in average velocity in the experiment (0.732) being higher than predicted by CFD (0.648), which corresponds to an error of -11.5%.

Overall, the results obtained in this study seem to agree reasonably well with the experimental data. Where discrepancies with the experiment are observed, comparison with other CFD participants shows that results are consistent between them. Such behaviour might indicate a common limitation of CFD as a method or of the experimental campaign, rather than errors of this study. The source of such discrepancy, however, is difficult to establish.

4.2.2 Combined rotation and surge

Next, a case with both rotor rotation and surge is validated against available data.

Figure 16 shows the variation of thrust and torque on the rotor with time. Additionally, x-displacement of the rotor due to surge and corresponding x-velocity are plotted on the same graph. Positive values of x-displacement correspond to the turbine moving



(a) along-wake **(b)** cross-wake

Figure 15: Downstream velocity distribution (fixed isolated rotor)

downstream, out of the wind, while negative values of x-displacement correspond to the turbine moving upstream, to the wind. Similarly, the positive values of x-velocity mean that the turbine is moving downwind, thus reducing the total inflow speed on the rotor, while the negative values of x-velocity mean that the turbine is moving upwind, which results in an increase of the apparent wind speed.

The rotor performance is observed to oscillate in phase with surge velocity. When the velocity peaks (i.e., the apparent wind is the smallest), both thrust and torque reach their corresponding minimum values. Conversely, when the velocity is negative (i.e., the apparent wind speed is the highest), the rotor experiences maximum loading. This behaviour was expected.

What is unexpected, however, is the mean value of the power coefficient, which is lower for the case of a surging wind turbine than in case of a fixed rotor. On the other hand,

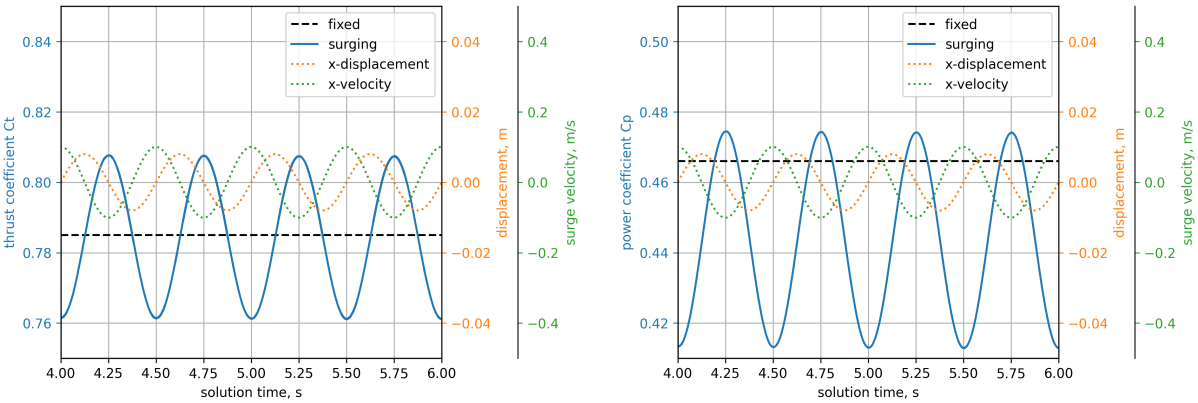


Figure 16: Variation of thrust and torque on a surging isolated rotor

mean thrust is the same for both surging and fixed rotors. It has not been identified why thrust and torque behaved differently in that regard.

As for comparison with other available data, figure 17 depicts experimental measurements over one surge cycle. The measurement signal is quite noisy and is dominated by the rotor excitation frequency P1 at 4Hz. As a result, it is difficult to make direct comparison regarding time evolution of the results. The same figure also illustrates measurements of other CFD participants, and the qualitative agreement is good.

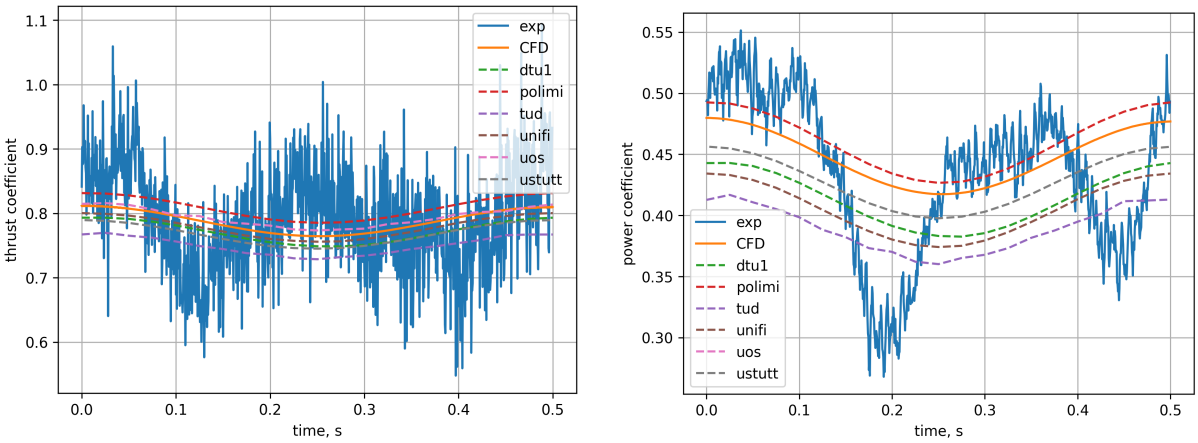
As for the numerical values, table 8 lists thrust and torque amplitudes experienced by the rotor. To extract the experimental values from the measurements, the power spectral density (PSD) of the signal was estimated using the `signal.welch()` function of the `SciPy` module in Python. The PSD at the surge frequency (2Hz) was converted to the corresponding amplitude and assumed to be equal to the amplitude of the quantity of interest.

Overall, despite a relatively large error in mean values (see table 7), the relative amplitudes due to surge were predicted at higher accuracy. It also appears that torque is more sensitive to variation in apparent wind speed, as its amplitude reaches 6.4% of the mean value, while thrust varies only by 2.5%.

Next, velocity profiles in the wake are compared. Figure 18 shows velocity distribution obtained in the experiment and in this study for both steady and unsteady cases. Here, a clear contradiction between the results is evident. According to the experiment, introduction of the surging motion does not affect velocity distribution behind the turbine significantly. Similar conclusions were drawn from the results of other CFD participants (see figures in Appendix F). Contrary to that, CFD results obtained in this project show that velocities in the wake are different when the rotor oscillates.

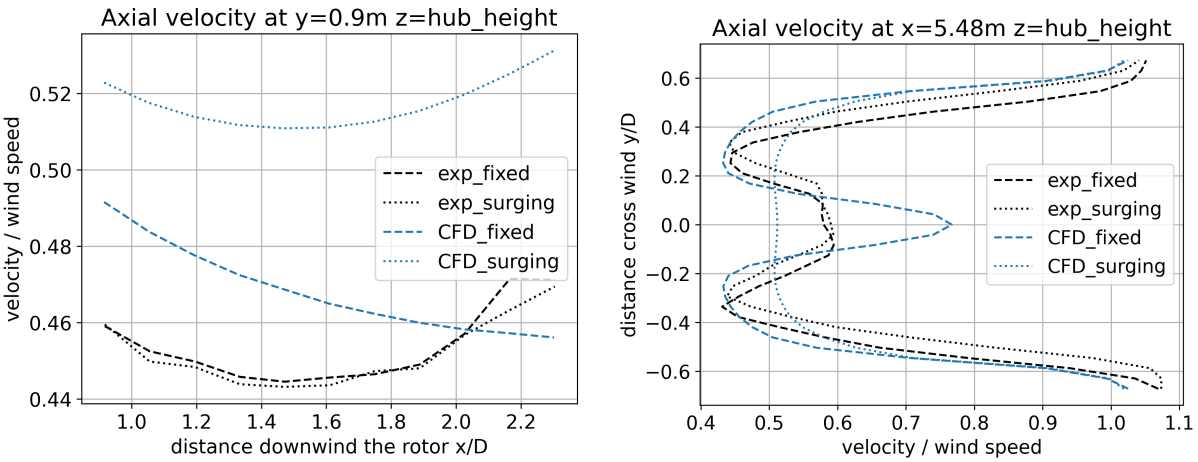
Table 8: Comparison with the experiment (surging isolated rotor)

Parameter	Experiment	CFD	
	value, -	value, -	difference, %
mean thrust coefficient	0.798	0.785	-1.6
thrust coefficient amplitude	0.016	0.023	30.4
relative thrust coefficient amplitude	2.0	2.9	0.9
mean power coefficient	0.433	0.444	2.5
power coefficient amplitude	0.038	0.031	-18.4
relative power coefficient amplitude	8.8	6.9	-1.9



(a) Thrust **(b)** Torque

Figure 17: Comparison of thrust and torque with the experimental data (surging isolated rotor)



(a) along-wake **(b)** cross-wake

Figure 18: Downstream velocity distribution (surging isolated rotor)

The wakes for both cases (steady and surging) are visualised in figure 19. Firstly, it is immediately apparent that the wake directly behind the hub is more uniform behind the surging rotor than behind the steady rotor. Perhaps, this is due to more mixing experienced by the blade root vortex and its consequent faster breakdown. Presumably, less intensive root vortex allows the wake confined within the tip vortices to recover quicker, thus resulting in a more uniform profile depicted in figure 18. The wakes can be characterised by the average speed of 0.685, which is 5.7% higher than in the fixed case (0.648).

In sum, although rotor forces were predicted well for an unsteady rotor, wake velocities saw a considerable deviation from the experiment and other CFD participants. At the same time, the evidence of faster wake recovery does agree with conclusions of H. Lee and D.-J. Lee (2019) and Rezaeiha and Micallef (2021). Establishing the reasons behind the discrepancy between experimental and numerical results is not straightforward and, perhaps, requires a more thorough investigation of experimental and numerical setups of different participants.

4.3 Turbine in a wake

The next test cases involved simulating steady and surging turbines in their own wake. Figure 20 shows time histories of thrust and torque measurements on steady and surging, isolated and in wake, rotors. The flow is also visualised in figures 21 and 22 for fixed and surging cases respectively.

It is immediately apparent that rotor performance starts to drop at around $t = 4s$, which is the expected behaviour as this is approximately how long it should take the wake to travel the length of the computational domain (16 m) at the speed of the oncoming wind (4.19 m/s). The change in thrust and power coefficients seems to be quite different between fixed and surging rotors. Table 9 lists mean thrust and power coefficients averaged over one rotation period in case of a steady rotor. Thrust and torque decreased by 54% and 90% respectively. The drop in performance appears to be overpredicted due to a plume of very low speed which formed at the very beginning of the simulation and was passing through the rotor when the final measurements were recorded (see figure 21). Most likely, rotor thrust and torque will recover to assume higher values once the plume passes.

Table 10 gives results for the surging cases, with mean values averaged over one last surge cycle and amplitudes taken for one last cycle only. This time, the drop in thrust

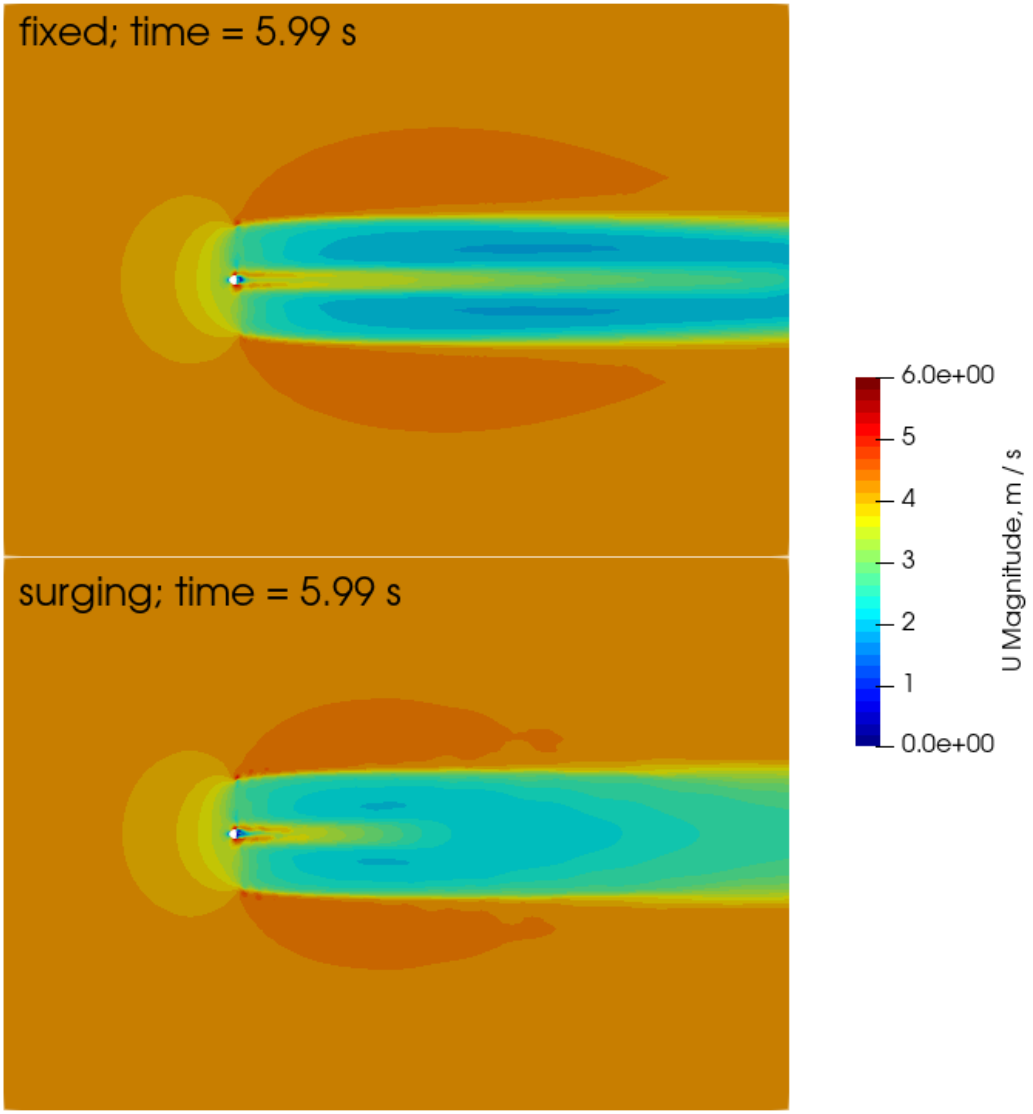
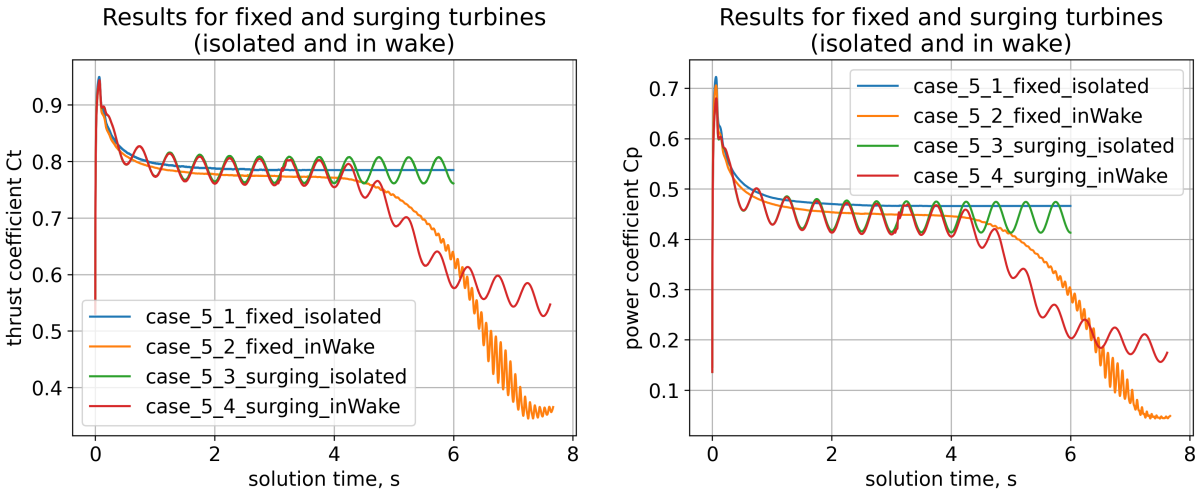


Figure 19: Wake visualisation for a steady (top) and a surging (bottom) turbine



(a) thrust

(b) torque

Figure 20: Comparison of thrust and torque of fixed and surging, isolated and in wake, rotors

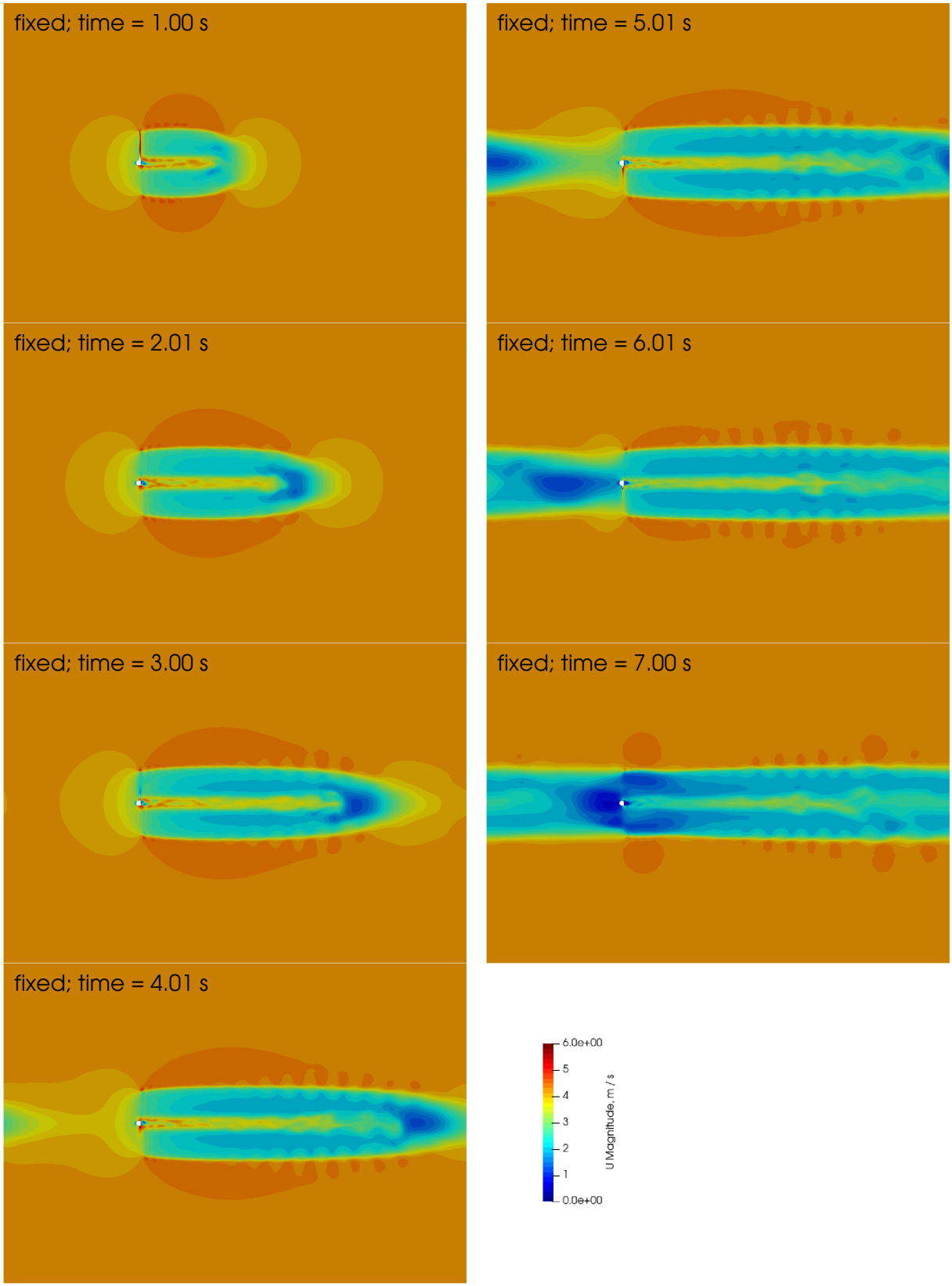


Figure 21: Wake propagation behind a steady rotor

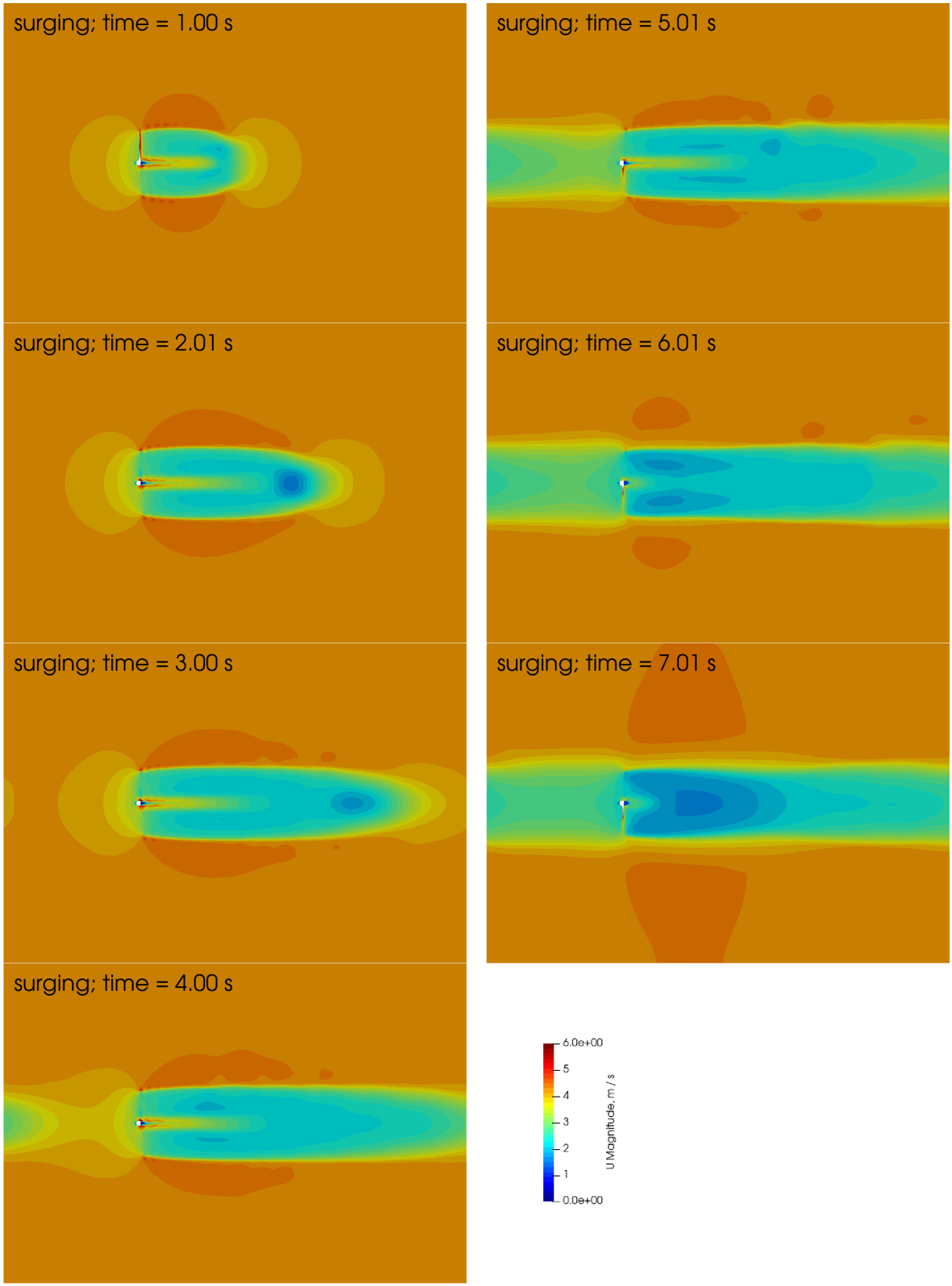


Figure 22: Wake propagation behind a surging rotor

Table 9: Comparison between an isolated turbine and a turbine in wake (steady cases)

Case	Mean thrust coefficient		Mean power coefficient	
	value, -	difference, %	value, -	difference, %
steady, isolated	0.785	-	0.466	-
steady, in wake	0.359	-54.3	0.048	-89.7

and power coefficients were 28% and 57% respectively, which is lower than for the fixed rotor. Such results were expected, considering that average wind speed in wake (estimated in section 4.2.2) was higher behind a surging rotor than behind a fixed one. Although, allowing the fixed rotor to go through the plume of low wind speed first would have allowed to make a more fair comparison.

The amplitudes of thrust and torque decreased by 9.5% and 9.7% respectively. This relatively modest change in amplitudes might be explained by the fact that motion parameters remained the same, meaning the change in the apparent wind speed was the same, in absolute terms, between the cases. Perhaps, such scenario might be not representative of the actual floater response, because the change in forces generated by the rotor will most likely affect the overall motion parameters of the structure.

Another observation of the behaviour of the rotor integral parameters, which can be drawn from figure 20, is the presence of high frequency oscillations on the fixed rotor, which are not visible on the surging rotor. This might indicate that the wake behind the fixed rotor has more high-energy vortices within it, which persist all the way to the 2nd-row turbines. The surging-rotor wake, on the other hand, is more uniform due to more intensive mixing and consequent breakdown of big vortices. This point can also be supported by comparing figures 21 and 22.

Overall, it appears that surging motion of a rotor leads to a more uniform wake structure with higher average velocity than when a rotor is fixed. Consequently, a higher

Table 10: Comparison between an isolated turbine and a turbine in wake (unsteady cases)

Case	Thrust coefficient		Power coefficient	
	value, -	difference, %	value, -	difference, %
surging, isolated, mean	0.784	-	0.444	-
surging, in wake, mean	0.556	-28.0	0.183	-57.0
surging, isolated, amplitude	0.023	-	0.031	-
surging, in wake, amplitude	0.021	-9.5	0.028	-9.7

power coefficient was observed on a surging rotor, when the spacing between rotors is kept the same between cases. Furthermore, the surging rotor did not experience higher-frequency oscillations, which might reduce its fatigue loading.

Results discussed in this subsection indicate the performance of a turbine in the second row. Continuing the simulations would allow to model the turbines in the subsequent rows. Unfortunately, due to time limitations, they were not run for any longer.

5 Conclusion

This work investigated the influence of surging motion on performance of a wind turbine rotor using computational fluid dynamics. Investigated test cases included an isolated fixed and an isolated surging rotor, as well as fixed and surging rotors in wake. The rotor was simulated explicitly using the available 3D CAD geometry, while turbulence was modelled with the $k-\omega$ SST model. To account for rotor rotation, sliding mesh interface was used, while surge was introduced by means of mesh morphing. A turbine in a wake was modelled using a cyclic boundary condition at domain outlet and inlet. Rotor integral forces and velocity distribution downwind the turbine were monitored. The steady test cases were validated against available experimental data from the UNAFLOW campaign. The following conclusions were drawn from the results.

- From the mesh convergence study (see section 4.1) it was determined that thrust readings converged, while torque readings did not. Further refinement of the mesh would be desirable but increased computational time would not have allowed to complete the project in the given time frame. As a result, a relatively coarse mesh was used.

- Comparison with experiments of a fixed rotor (see section 4.2.1) showed good agreement for thrust (-1.5%), but worse agreement for torque (7.8%), which was expected following the results of the convergence studies. Wake velocity profiles agreed well (RMS of the error 4.8% for along-wake and 15.8% for cross-wake). The cross-wake velocity agreed better with other available CFD results, which lead to believe that the source of discrepancy was due to wind tunnel wall blockage, which constrained the wake evolution.

- Comparison with experiments of a surging rotor (see section 4.2.2) saw bigger differences. On the one hand, relative amplitudes of thrust and torque were within 2% of each other. On the other hand, wake velocities were clearly different. Experimental results suggested that, for the used parameters, rotor movement has negligible effect on wake velocity deficit, whereas obtained CFD results suggest that wake velocities were higher by about 5% in case of a surging rotor. No clear consensus on the matter exists in the literature, as some works claim that rotor movement aids mixing and therefore faster wake recovery (H. Lee and D.-J. Lee, 2019; Rezaeiha and Micallef, 2021), while others have not observed such phenomena (Bergua et al., 2022; Cioni et al., 2023).

- Simulations of turbines in wake (see section 4.3) seem to indicate that the surging motion leads to more intensive mixing in the wake, which promotes faster breakdown of vortical structures and a more uniform velocity field. As a consequence, power output of a downwind turbine increases in comparison to when the upwind rotor is fixed.

In summary, the obtained results suggest that, due to faster wake recovery, floating offshore wind turbines in farms might be spaced at smaller distances, perhaps reducing balance of plant and maintenance costs. Conversely, keeping the turbines at already established distances might lead to higher energy yield than for fixed turbines.

Regarding the future works, it would be beneficial to further investigate the cause for discrepancy in wake velocity predictions by the UNAFLOW experiment and the CFD results. Additionally, a wider range of input parameters, such as surge amplitude and frequency, should be investigated to assess rotor performance in more extreme conditions.

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A Mesh generation controls

Table A1 lists mesh generation controls specified in the `meshDict` dictionary for the `cfMesh` utility. Note that whole numbers refer to the `refinemntLevel` setting, while real numbers refer to the `cellThickness` setting.

Complete specifications can be found in a GitHub repository `https://github.com/alextsvet99/mscp_cases`.

Table A1: Mesh generation controls

Region (surface)	Units	Value
rotating region		
`maxCellSize`	m	0.032
`rotor`	-	4
`tip`	-	6
`trailing_edge`	-	6
`leading_edge`	-	6
`te_bot`	-	5
static region		
`maxCellSize`	m	0.256
`ami`	-	3
`wake`	-	2
`root_vortex`	-	3
`tip_vortex`	-	3

B OpenFOAM dictionaries listings

All test cases files are available in a GitHub repository `https://github.com/alextsvet99/mscp_cases`.

B.1 fvSchemes

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  v2106                                 |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      fvSchemes;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

ddtSchemes
{
    default         Euler;
}

gradSchemes
{
    default         leastSquares;
}

divSchemes
{
    default         none;
    div(phi,U)      Gauss linearUpwindV grad(U);
```

```
    div(phi,k)      Gauss linear;
    div(phi,omega)  Gauss linear;
    div((nuEff*dev2(T(grad(U))))) Gauss linear;
}

laplacianSchemes
{
    default         Gauss linear corrected;
}

interpolationSchemes
{
    default         linear;
}

snGradSchemes
{
    default         corrected;
}

wallDist
{
    method          meshWave;
}

// ************************************************************************* //
```

B.2 fvSolution

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  v2106                                 |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      fvSolution;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

solvers
{
    "pcorr.*"
    {
        solver          GAMG;
        tolerance       1e-2;
        relTol          0;
        smoother        DICGaussSeidel;
        cacheAgglomeration no;
        maxIter         50;
    }

    p
    {
        $pcorr;
        tolerance       1e-5;
        relTol          0.01;
    }
```

```
    pFinal
    {
        $p;
        tolerance       1e-6;
        relTol          0;
    }

    "(U|k|omega)"
    {
        solver          smoothSolver;
        smoother        symGaussSeidel;
        tolerance       1e-6;
        relTol          0.1;
    }

    "(U|k|omega)Final"
    {
        solver          smoothSolver;
        smoother        symGaussSeidel;
        tolerance       1e-6;
        relTol          0;
    }

    "(cellDisplacement|cellDisplacementFinal)"
    {
        solver          GAMG;
        tolerance       1e-6;
        relTol          0;
        smoother        GaussSeidel;
        cacheAgglomeration true;
        nCellsInCoarsestLevel 10;
        agglomerator    faceAreaPair;
        mergeLevels     1;
    }
}

PIMPLE
```

```
{
    momentumPredictor   yes;
    correctPhi          yes;
    nOuterCorrectors    50;
    nCorrectors         2;
    nNonOrthogonalCorrectors 1;
    pRefPoint (11.9 0 5.9);
    pRefValue 0;

    residualControl
    {
        p
        {
            tolerance       1e-4;
            relTol          0;
        }

        U
        {
            tolerance       1e-5;
            relTol          0;
        }
    }
}

relaxationFactors
{
    "(U|k|omega).*"   1;
}

cache
{
    grad(U);
}

// ************************************************************************* //
```

B.3 dynamicMeshDict

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  v2106                                 |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      dynamicMeshDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dynamicFvMesh dynamicMultiMotionSolverFvMesh;

dynamicMultiMotionSolverFvMeshCoeffs
{
    turbine
    {
        motionSolverLibs (fvMotionSolvers);
        motionSolver     solidBody;
        cellZone         rotating_region;
        solidBodyMotionFunction  multiMotion;

        rotation
        {
            solidBodyMotionFunction rotatingMotion;
            origin (0 0 0);
            axis (1 0 0);
            omega 25.23746; // rad/s
        }
```

```
        surge
        {
            solidBodyMotionFunction oscillatingLinearMotion;
            amplitude (0.008 0 0);
            omega 12.56637; // rad/s;
        }
    }

    outer_domain
    {
        motionSolverLibs (fvMotionSolvers);
        motionSolver displacementLaplacian;
        cellZone stationary_region;
        diffusivity exponential 2 inverseDistance (ami_out);
    }
};

// ************************************************************************* //
```

C Boundary and initial conditions

Tables C1 and C2 list boundary and initial conditions. Please note that for simulations with cyclic inlet and outlet, boundary conditions for patches `inlet` and `outlet` should be changed to `cyclicAMI` in table C1 and to `patchType cyclicAMI type fixedValue value uniform (0 0 0)` in table C2.

Example test cases can be found in a GitHub repository `https://github.com/alextsvet99/mscp_cases`.

Table C1: Boundary and initial conditions required for all cases

Patch \Variable	U	p	k	omega	nut
internalField	uniform (4.19 0 0)	uniform 0	uniform 0.01	uniform 1.60	uniform 0
inlet	fixedValue uniform (4.19 0 0)	zeroGradient	fixedValue $internalField	fixedValue $internalField	calculated uniform 0
outlet	inletOutlet uniform (0 0 0) uniform (0 0 0)	fixedValue unifrom 0	inletOutlet $internalField $internalField	inletOutlet $internalField $internalField	calculated uniform 0
sides	slip	zeroGradient	zeroGradient	zeroGradient	zeroGradient
rotor	movingWallVelocity uniform (0 0 0)	zeroGradient	kqRWallFunction uniform 0	omegaWallFunction uniform 1460	nutkWallFunction uniform 0
ami_in	cyclicAMI	cyclicAMI	cyclicAMI	cyclicAMI	cyclicAMI
ami_out	cyclicAMI	cyclicAMI	cyclicAMI	cyclicAMI	cyclicAMI

Table C2: Boundary and initial conditions required for unsteady cases

Patch \Variable	pointDisplacement	cellDisplacement
internalField	uniform (0 0 0)	uniform (0 0 0)
inlet	fixedValue uniform (0 0 0)	fixedValue uniform (0 0 0)
outlet	fixedValue uniform (0 0 0)	fixedValue uniform (0 0 0)
sides	fixedValue uniform (0 0 0)	fixedValue uniform (0 0 0)
rotor	fixedValue uniform (0 0 0)	cellMotion uniform (0 0 0)
ami_in	patchType cyclicAMI; type fixedValue; value uniform (0 0 0);	patchType cyclicAMI; type cellMotion; value uniform (0 0 0);
ami_out	patchType cyclicAMI; type solidBodyMotionDisplacement; solidBodyMotionFunction oscillatingLinearMotion; amplitude (0.008 0 0); omega 12.56637; // rad/s;	patchType cyclicAMI; type cellMotion; value uniform (0 0 0);

D Mathematical description of the studied parameters

The following formulae were used to calculate the values for the studied parameters.

Thrust force is expressed as a thrust coefficient C_t:

$$C_t = \frac{T}{0.5\rho V^2 A}, \tag{6}$$

where $T,\ N$ is the measured thrust force; $\rho,\ kgm^{-3}$ is the air density; $V,\ ms^{-1}$ is the far-field wind speed; $A = \pi R^2,\ m^2$ is the characteristic area, in this case the rotor swept area of rotor radius $R,\ m$.

Torque is expressed in terms of power coefficient C_p:

$$C_p = \frac{P}{0.5\rho V^3 A}, \tag{7}$$

where $P = 2\pi nQ,\ W$ is the power; $n,\ s^{-1}$ is the rotor speed; $Q,\ Nm$ is the measured rotor torque.

Linear dimensions are normalised with respect to the rotor diameter $D = 2R = 2.37\ m$, while velocities - by the far-field wind speed $V = 4.19\ ms - 1$.

The error in the results is calculated as:

$$e = \frac{cfd - exp}{exp}, \tag{8}$$

where cfd is the value obtained in CFD simulations, while exp is the experimental value.

In case of velocity profiles, the error is further characterised by computing its root mean square (RMS) value:

$$RMS = \sqrt{\sum e^2/n}, \tag{9}$$

where the error e is calculated using eq. 8 for each of the n points in the profile.

An average velocity in the wake u_{avg} is calculated from the cross-wake velocity profiles using integral averaging over available measurement points:

$$u_{avg} = \frac{1}{area}\int_A udA = \frac{1}{area}\int_A u\pi dr^2 \approx \frac{1}{\pi r_n^2}\sum_{i=0}^{n-1}\frac{u_{i+1}+u_i}{2}\cdot\frac{\pi|r_{i+1}^2 - r_i^2|}{2} \tag{10}$$

where $area$ is the circular area covered by the available data; u is the velocity at a point in wake; r is the radius from hub line to the point; summation is performed over n semicircles of radii r_i with corresponding velocities u_i.

E Estimation of discretisation error

E.1 Methodology

A discretisation error will be attempted to be estimated using an approach outlined in Versteeg and Malalasekera (2007). The error e is assumed to be described by the function:

$$e(h) = U_{exact} - U_{cfd} \approx ch^p, \tag{11}$$

where U_{exact} is the hypothetical exact value of the variable of interest; U_{cfd} is the value obtained in the CFD simulation; c is the constant of proportionality; h is the reference mesh size; p is the order of the numerical scheme, i.e., the power that governs the rate at which the error tends to zero as the grid is refined.

Running a number of simulations at different mesh sizes h will allow to obtain corresponding U values, from which the constants in equation 11 can be evaluated. Assuming $U_1,\ U_2,\ U_3$ to be the results on the meshes of decreasing sizes $h_1,\ h_2,\ h_3$, which are related by a constant refinement ratio $r = h_1/h_2 = h_2/h_3$, the observed order p_{obs} can be calculated:

$$p_{obs} = \log \frac{U_2 - U_1}{U_3 - U_2} / \log r \tag{12}$$

The constant of proportionality can be obtained as:

$$c = \frac{U_2 - U_1}{h_1^{p_{obs}} - h_2^{p_{obs}}} \tag{13}$$

The final value of c will be the assumed to equal the average of values obtained for two sets of results (U_2 and U_1, U_3 and U_2).

A complete investigation of error and uncertainty of CFD results requires further assessment of parameters such as round-off error, iterative convergence error, and input uncertainty (e.g. initial conditions of turbulence properties) (Versteeg and Malalasekera, 2007). However, it will not be carried out in this project. Nevertheless, it will be interesting to compare the error predicted by the outlined method and the the actual error with respect to the experimental results.

E.2 Results

To estimate the error introduced by the spatial discretisation, the procedure outlined above was used. Table E1 lists the calculated observed accuracy of the scheme p_{obs},

constant of proportionality c, and predicted error on the finest mesh e for both thrust and torque. The estimated values for the thrust coefficient do not appear to be reasonable as the order of the observed scheme should not exceed 2. This results might suggest that the error in this region does not decrease asymptotically, which renders the assumption of the error distribution invalid. This result might be expected as the relative difference between the readings is less than 0.5% (see table 5). The results for torque appear to be more sensible, as the observed accuracy of the scheme falls within the range $[0; 2]$. Although the FVM is nominally 2nd-order accurate, it is not uncommon for the accuracy to reduce to the 1st-order or even 0-th order (Syrakos et al., 2017). The predicted error on the finest grid relative to an exact solution is 7% and will be compared to the error obtained later in the validation section. Similar to the conclusion made above, this might indicate that convergence for torque has not been reached.

Table E1: Error estimation

Parameter	Thrust coefficient	Power coefficient
p_{obs}	15.3	1.11
c	0.00001	0.033
e, %	0.01	6.59

F Comparison of wake velocity profiles for fixed and surging cases

Figures F1 and F2 show along-wake and cross-wake velocity profiles for fixed and surging turbines, as obtained in the UNAFLOW experiment (denoted as "experiment"), in this study (denoted as CFD), and by a number of participant of the OC6 Phase III validation study (denoted as "dtu1", "polimi", and "unifi").

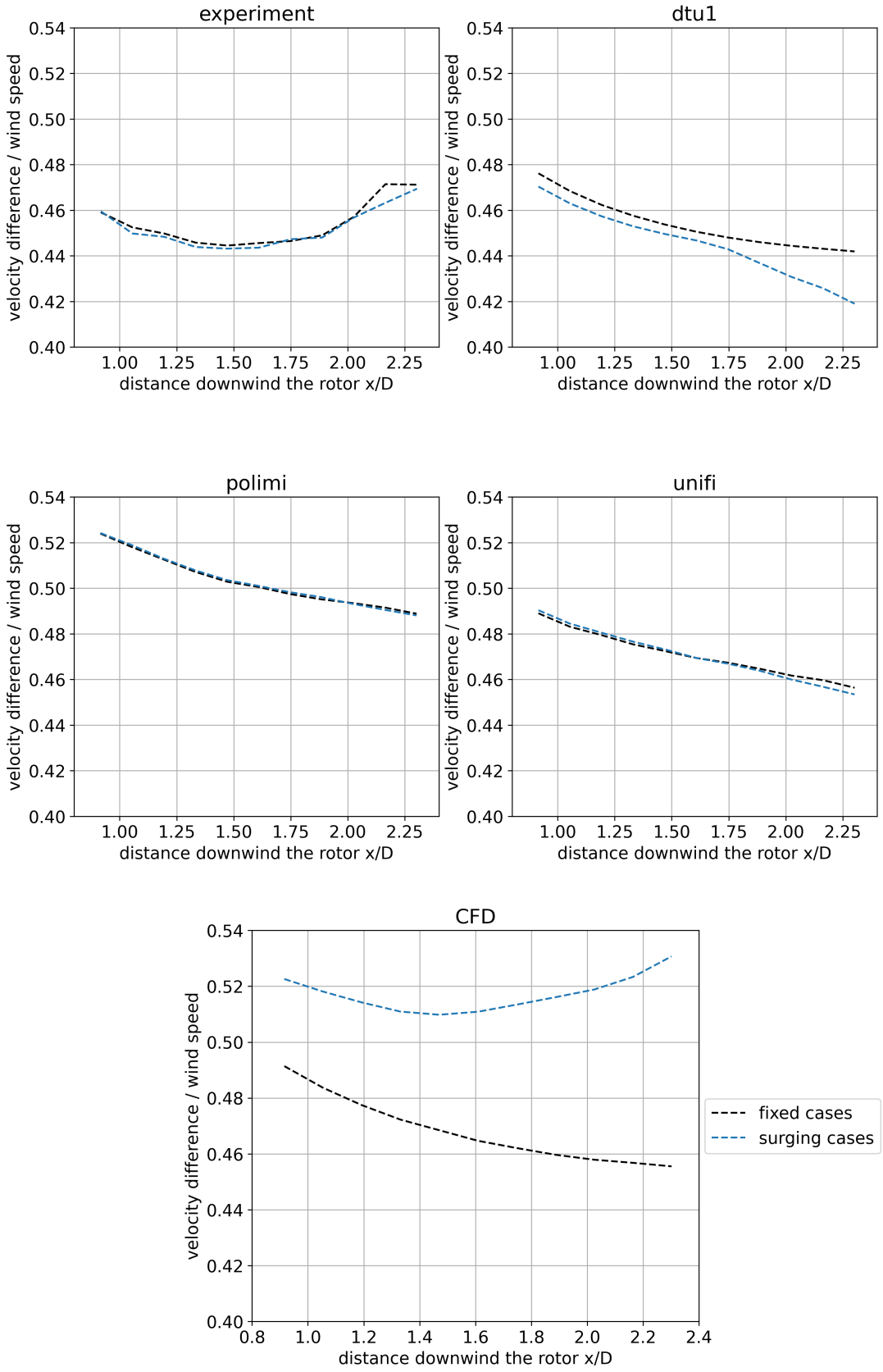


Figure F1: Along-wake velocity profiles for steady and surging rotors from various studies

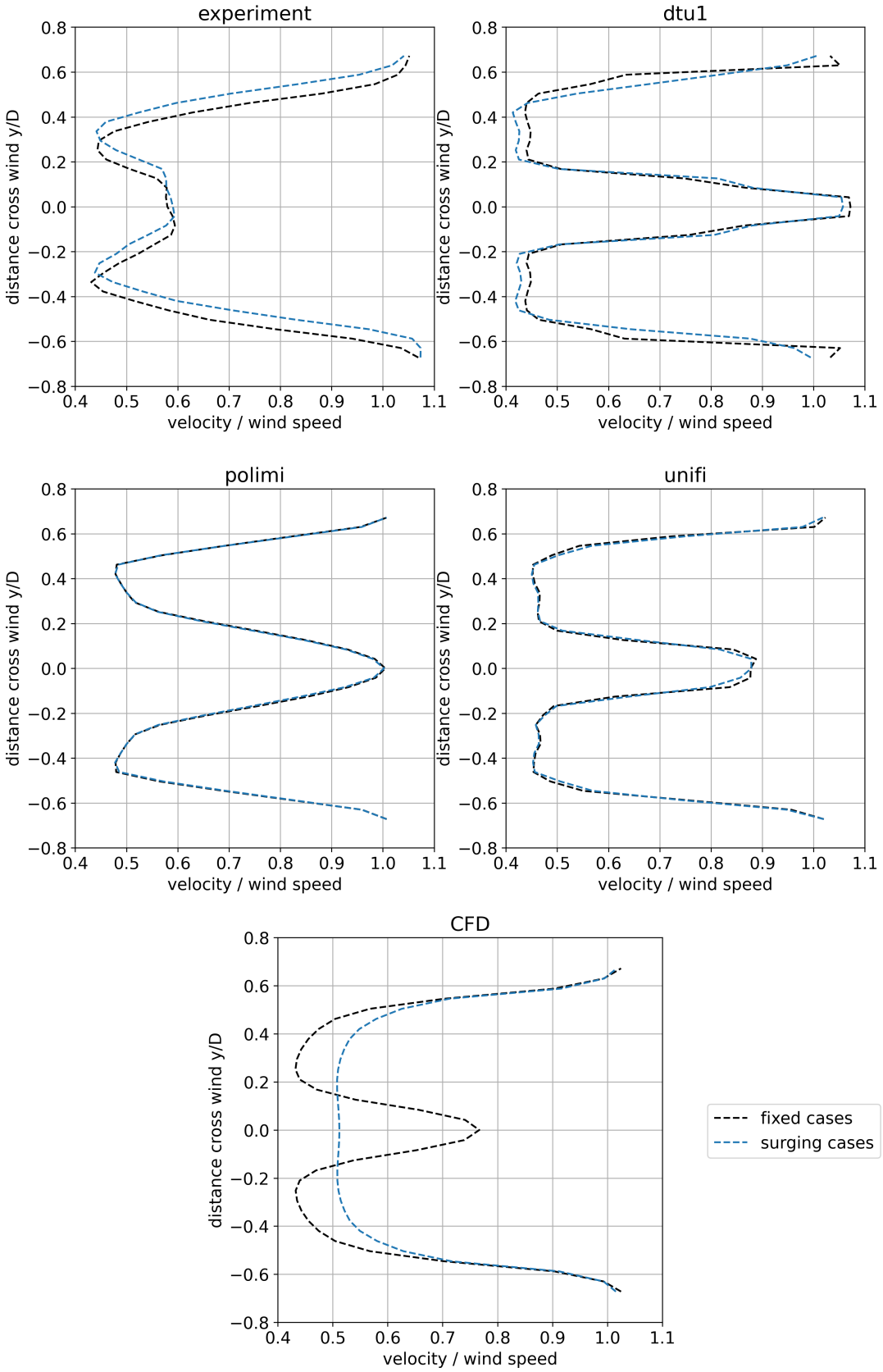


Figure F2: Cross-wake velocity profiles for steady and surging rotors from various studies